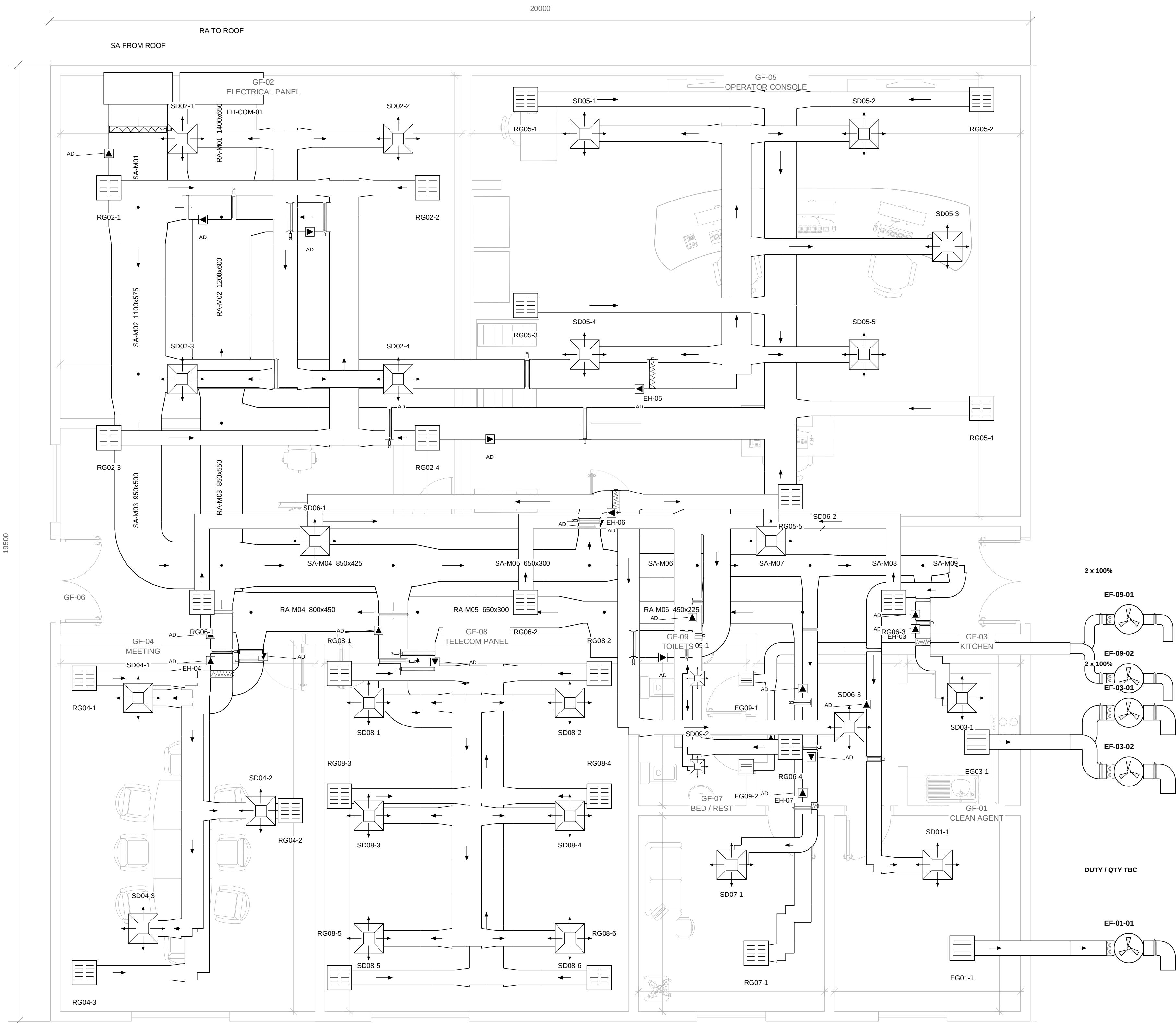




ROOM AIRFLOW SCHEDULE

GF	Room	SA L/s	RA L/s*	EA L/s*
02	ELECTRICAL PANEL	664.8	605.5	-
05	OPERATOR CONSOLE	777.7	718.4	-
04	MEETING	449.2	389.9	-
08	TELECOM PANEL	884.6	825.3	-
06	CORRIDOR	418.2	358.9	-
07	BED / REST	168.2	108.9	-
09	TOILETS	46.2	-	64.9
03	KITCHEN	147.6	-	147.6
01	CLEAN AGENT	141.3	-	TBC

DUCT / SYMBOL LEGEND

- SA - supply duct
- RA - return duct
- EA - dedicated extract
- SD - four-way supply diffuser
- RG - ducted return grille
- EG - extract grille
- VCD - balancing damper
- MFD* - conditional fire damper
- RCD - return control damper
- EH - inline electric heater
- AD - access door

Terminal aliases: SD02-1 = SD-02-01; RG / EG follow the same format. Full tags and quantities are in the registers.

Supply and return ducts crossing in plan are separate air paths. Actual duct elevations and insulation clearances require ceiling coordination.

* Return quantities are assumed balances. Extract references and +50 / +25 Pa targets require final airflow and pressure verification.

Units: mm / L/s / m/s. Terminal faces and clear duct widths are shown. Duct levels, selected fitting development and manufacturer clearances remain pending. No shared return from kitchen, toilet or clean-agent room.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COORDINATED DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-001

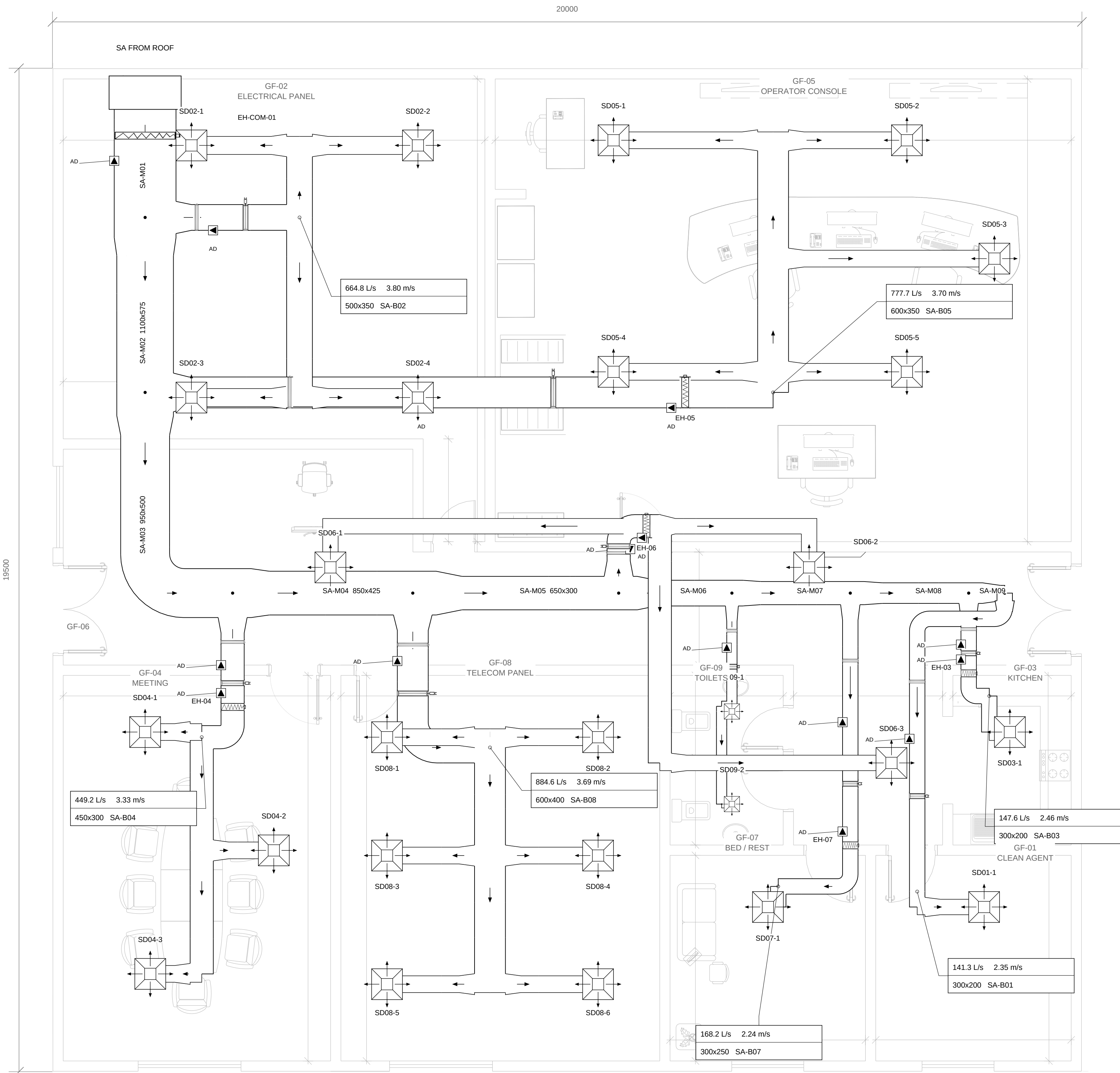
SCALE: 1:50 AT A1

REVISION

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07 OCT 2026

SHEET 1 OF 18



SUPPLY MAIN SECTIONS

Section	W x H mm	L/s	m/s
SA-M01	1200x650	3697.8	4.74
SA-M02	1100x575	3033.0	4.80
SA-M03	950x500	2255.3	4.75
SA-M04	850x425	1806.1	5.00
SA-M05	650x300	921.5	4.73
SA-M06	500x250	503.3	4.03
SA-M07	450x250	457.1	4.06
SA-M08	400x200	288.9	3.61
SA-M09	300x200	141.3	2.35

ROOM SUPPLY BRANCHES

GF	W x H mm	L/s	m/s
02	500x350	664.8	3.80
05	600x350	777.7	3.70
04	450x300	449.2	3.33
08	600x400	884.6	3.69
06	450x300	418.2	3.10
07	300x250	168.2	2.24
09	200x150	46.2	1.54
03	300x200	147.6	2.46
01	300x200	141.3	2.35

Room headers retain their branch size. Diffuser tags identify individual outlets. Round runout / neck diameters and each terminal airflow are listed on D-004 to D-006 and D-010.

MFD* is a conditional rated-penetration provision. VCD-Snn and FSD-Snn correspond to the GF-nn supply branch. Inline heaters are upstream of that room's terminal take-offs.

Continuous main tapers, flared branch boots and terminal adaptors: D-011. AD tags identify maintenance access doors. Proposed taper lengths and device coordinates are in the registers.

Total HAP-reference supply: 3697.80 L/s. Three 50% PAUs: two duty, rotating available alternate. Boxed callouts show branch flow, velocity and clear size. No room sensors are overlaid on this duct plan.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SUPPLY AIR DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-002

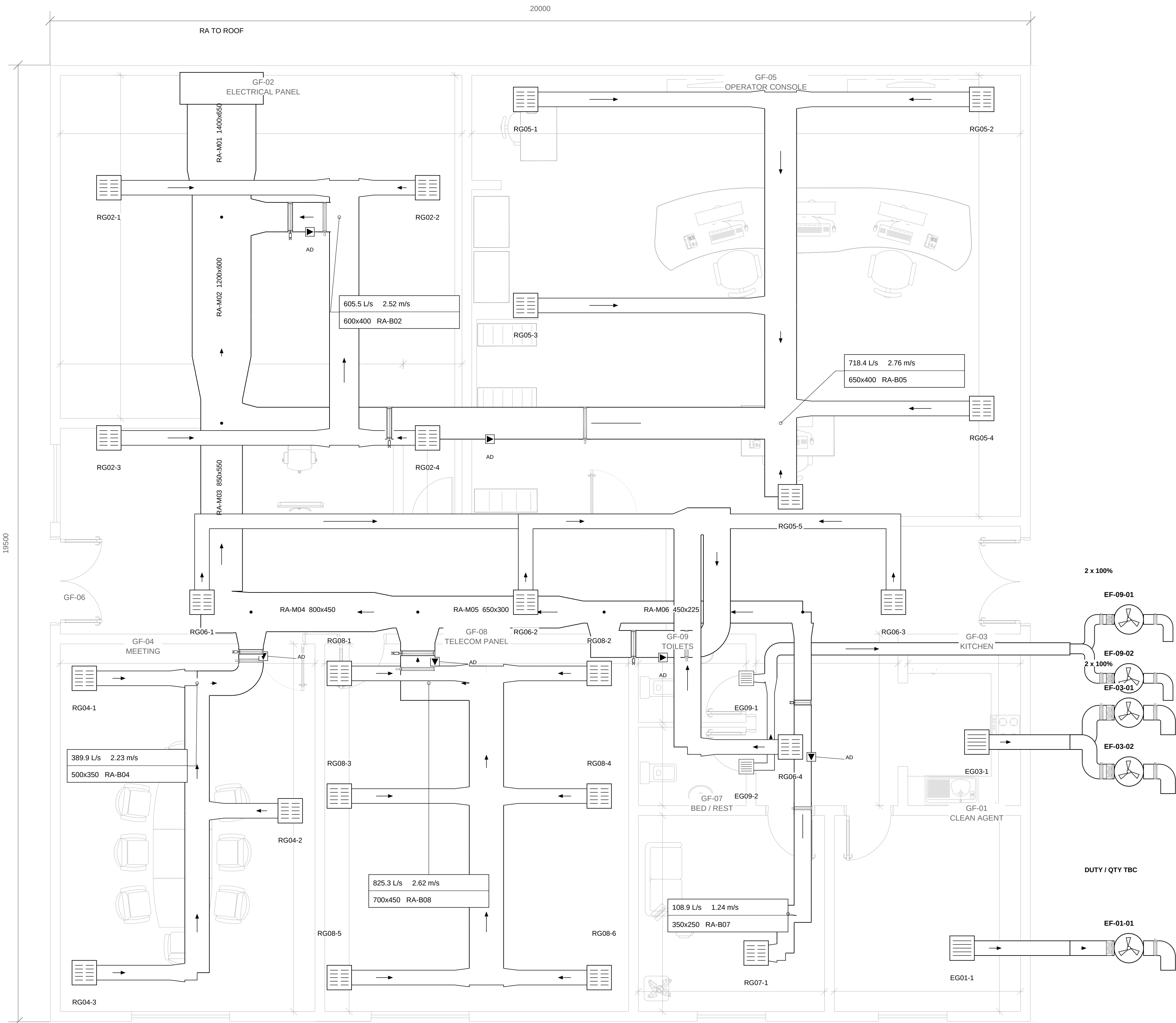
SCALE: 1:50 AT A1

REVISION

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SHEET 2 OF 18



RETURN MAIN SECTIONS

Section	W x H mm	L/s	m/s
RA-M01	1400x650	3006.7	3.30
RA-M02	1200x600	2401.2	3.34
RA-M03	850x550	1682.9	3.60
RA-M04	800x450	1293.0	3.59
RA-M05	650x300	467.7	2.40
RA-M06	450x225	108.9	1.08

ROOM RETURN / EXTRACT

GF	Air	W x H mm	L/s*
02	RA	600x400	605.5
05	RA	650x400	718.4
04	RA	500x350	389.9
08	RA	700x450	825.3
06	RA	550x350	358.9
07	RA	350x250	108.9
09	EA	250x200	64.9
03	EA	300x200	147.6
01	EA	TBC	TBC

No central return: GF-09 toilets, GF-03 kitchen and GF-01 clean-agent. Kitchen / toilet fans: 2 x 100% with duty rotation. CAG duty and quantity remain unverified.

Extract fan symbols are schematic equipment markers; actual fan and louver dimensions depend on manufacturer selection. Wall outlets face downwards.

* Return total 3006.68 L/s is an assumed starting balance. Existing extract references do not verify positive pressure or outdoor-air compliance. Confirmed fire shuts down all HVAC; automatic fire extraction is not enabled.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

RETURN AIR AND EXTRACT DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-003

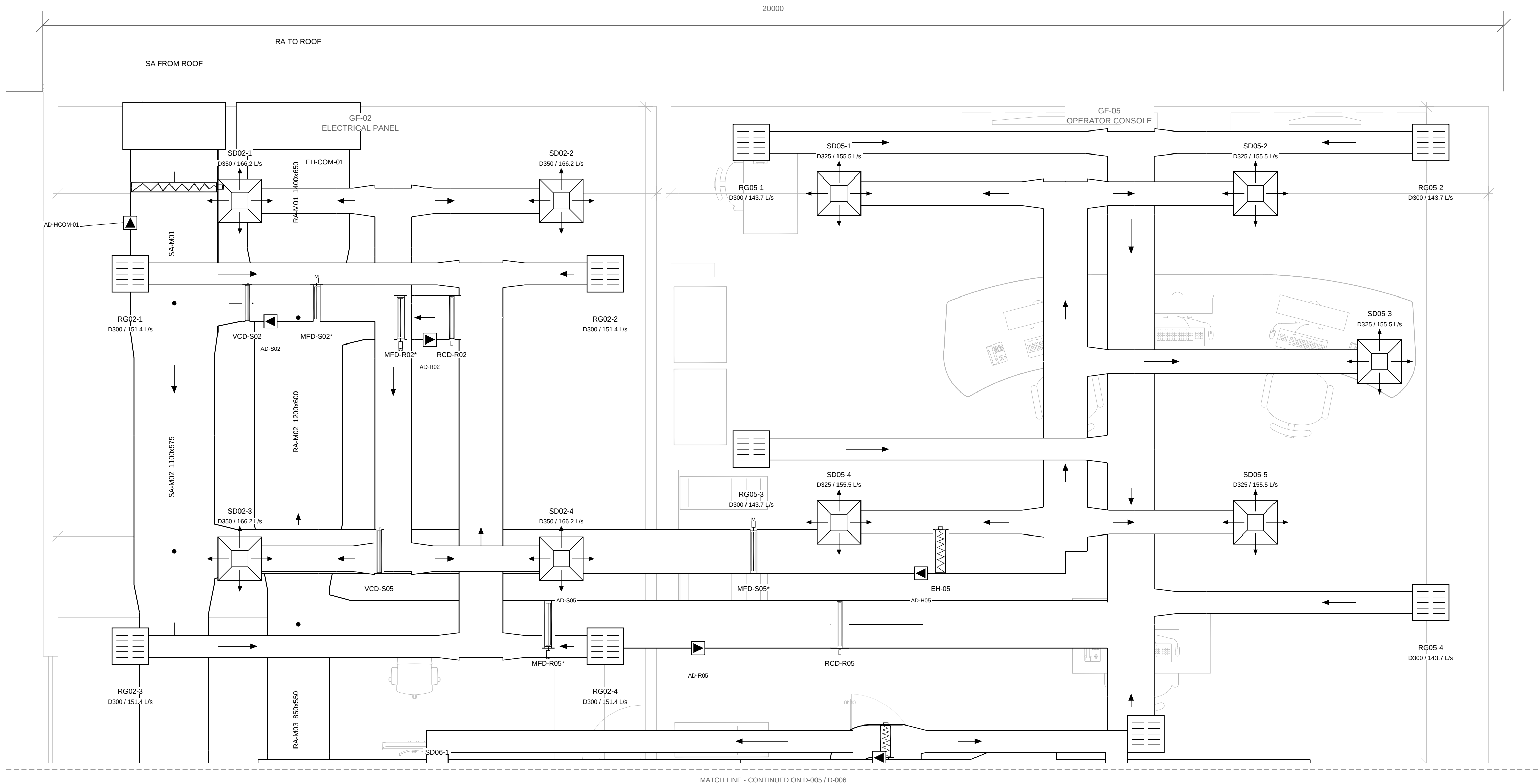
SCALE: 1:50 AT A1

REVISION

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SHEET 3 OF 18



NORTH-ROOM TERMINALS / BRANCHES

GF-02 ELECTRICAL PANEL

SA-B02 500x350 mm | 664.80 L/s | 3.80 m/s

RA-B02 600x400 mm | 605.46 L/s | 2.52 m/s

SD: 4 x 166.20 L/s; neck D350; face 600x600

RG: 4 x 151.37 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-05 OPERATOR CONSOLE

SA-B05 600x350 mm | 777.70 L/s | 3.70 m/s

RA-B05 650x400 mm | 718.36 L/s | 2.76 m/s

SD: 5 x 155.54 L/s; neck D325; face 600x600

RG: 5 x 143.67 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

Plan symbols identify VCD-S02 / S05, conditional MFD-S02 / S05*, RCD-R02 / R05 and conditional return MFDs. The component register retains the FSD tag prefix for the conditional motorized fire/smoke damper symbols.

Mains entering these rooms: SA-M01/M02/M03 and RA-M01/M02/M03. Source architectural orientation is retained. No structural grid, duct BOD, access envelope or manufacturer dimension is inferred from HAP room height.

Access doors: AD-Snn / AD-Rnn beside branch dampers; AD-Hnn beside heaters. Taper / access arrangements: D-011. Instrumentation: D-007 / D-012. Catalogue selection remains pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

NORTH ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-004

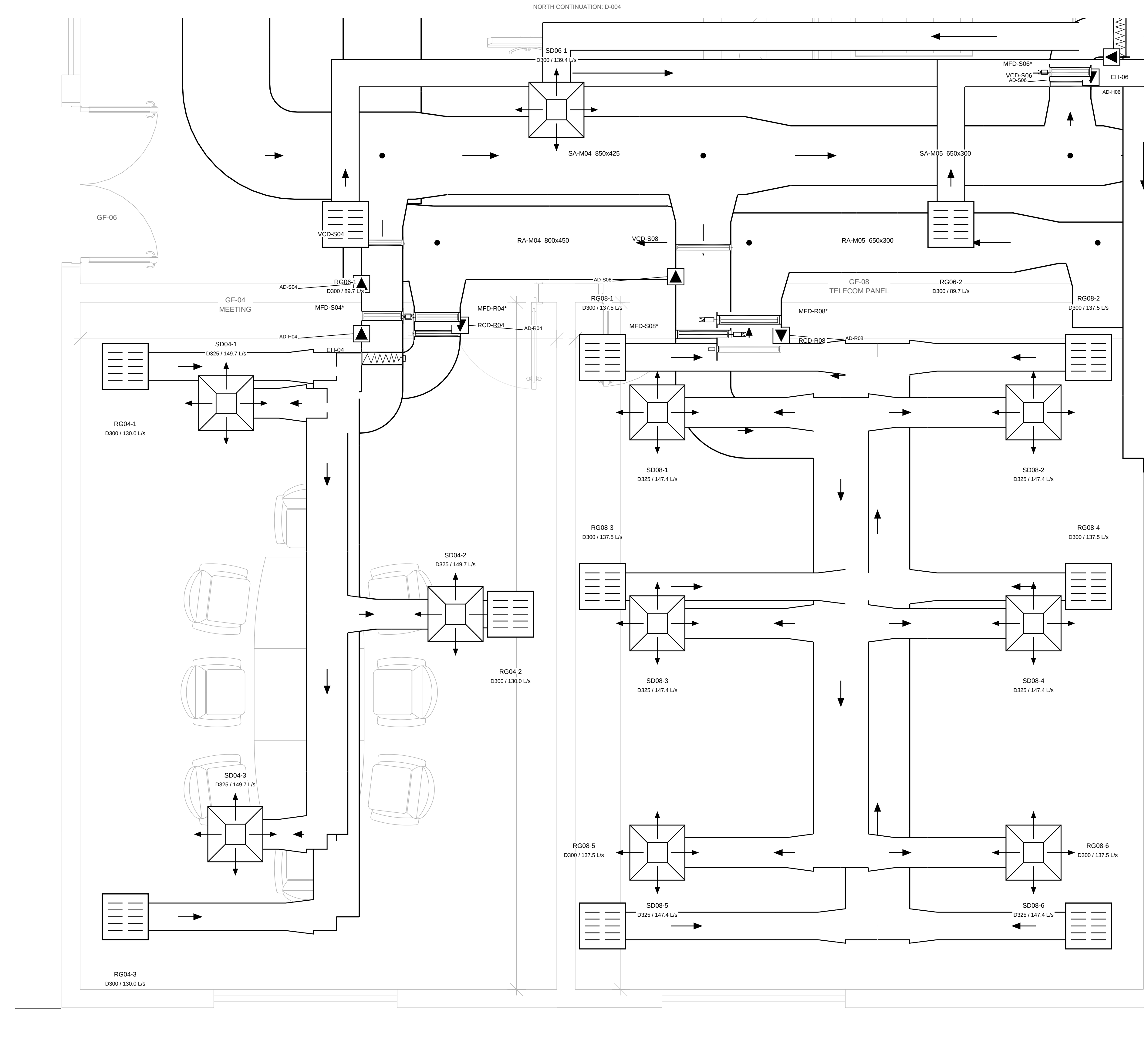
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REVISION

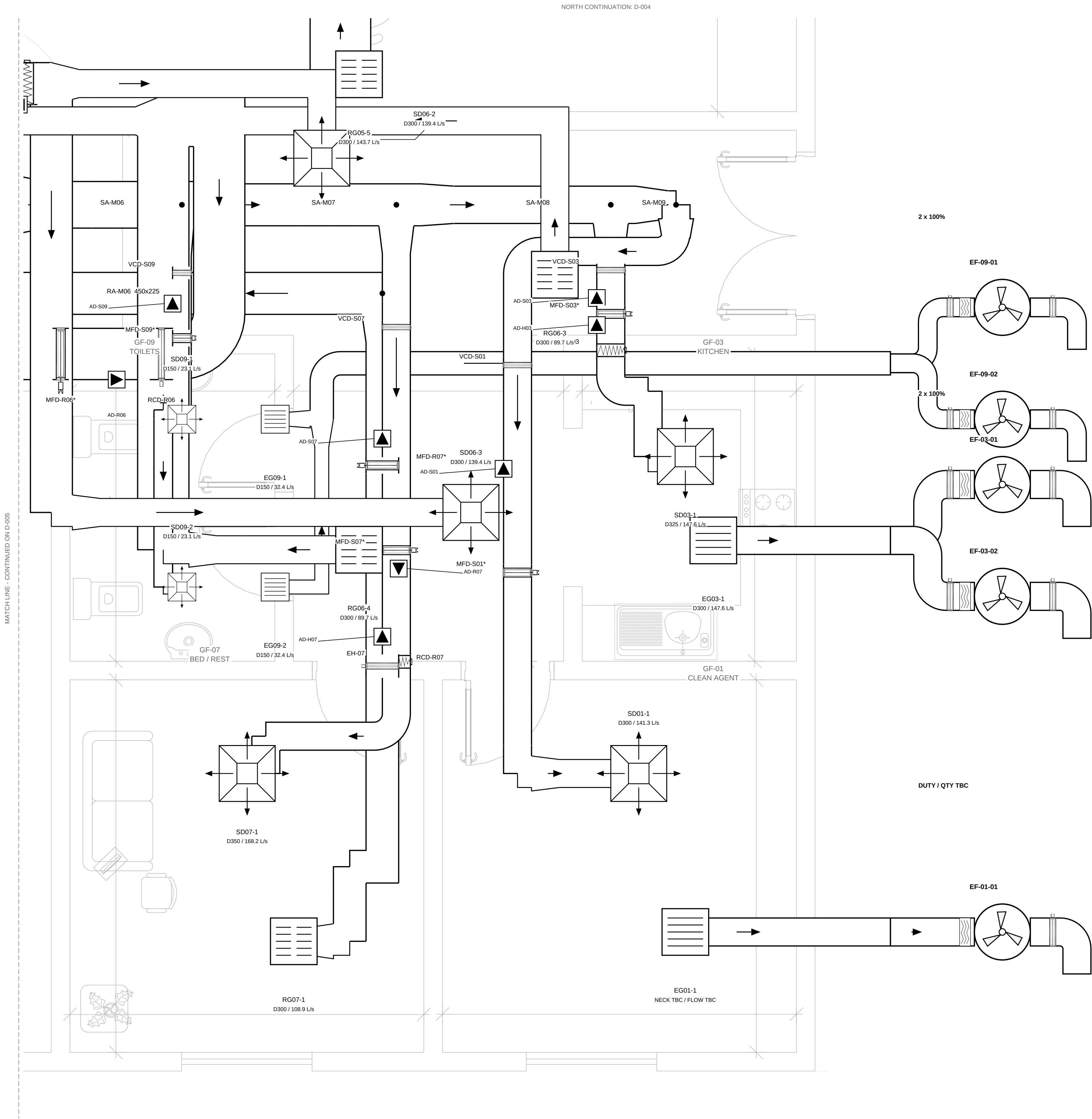
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GF-04 meeting; GF-08 telecom; GF-06 corridor. Room targets +50 Pa relative to outdoors; actual pressure verification remains pending. Full flow / velocity / horizontal route data is listed on D-010.



ROOM BRANCH / TERMINAL DATA

GF-09 TOILETS

SA-B09 200x150 mm | 46.20 L/s | 1.54 m/s

EA-B09 250x200 mm | 64.89 L/s | 1.30 m/s

SD: 2 x 23.10 L/s; neck D150; face 300x300

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-07 BED / REST

SA-B07 300x250 mm | 168.20 L/s | 2.24 m/s

RA-B07 350x250 mm | 108.86 L/s | 1.24 m/s

SD: 1 x 168.20 L/s; neck D350; face 600x600

RG: 1 x 108.86 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-03 KITCHEN

SA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

EA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

SD: 1 x 147.60 L/s; neck D325; face 600x600

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-01 CLEAN AGENT

SA-B01 300x200 mm | 141.30 L/s | 2.35 m/s

SD: 1 x 141.30 L/s; neck D300; face 600x600

EA / grille duty and pickup level: TBC

Room header: branch size. Terminal runout: listed neck.

EXTRACT ARRANGEMENTS

EF-09-01/02: toilets, 2 x 100%. EF-03-01/02: kitchen, 2 x 100%. EF-01-01: one CAG fan marker shown; final quantity, airflow and pickup elevation require confirmation.

Kitchen, toilet and CAG targets: +25 Pa to outdoors. Resting room: +50 Pa, with return air. Minimum ventilation: 10 ACH kitchen / toilet; 5 ACH CAG.

Toilet HAP supply is 46.2 L/s; minimum outdoor-air duty is not established by that mixed-air supply. Resolve the fresh-air duty before final HAP and pressure balance.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-EAST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-006

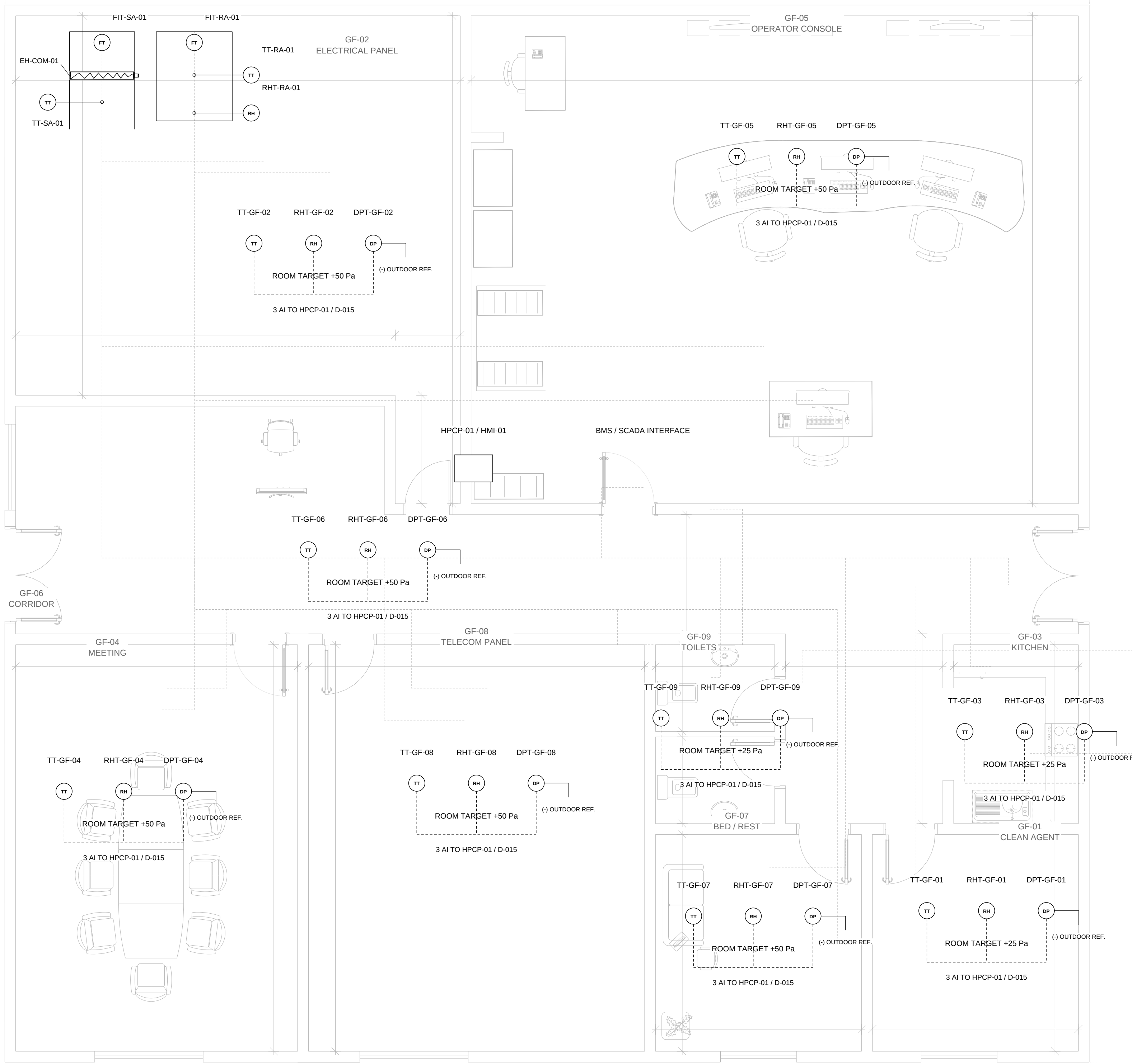
SCALE: 1:25 AT A1

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SHEET 6 OF 18



Room instrument locations are proposed. Plant/filter/fan instruments and heater AFS / TSHH protection: D-012. Final ranges, reference probes, mounting and routes require I&C coordination; operating sequence: S-004.

ROOM FIELD INSTRUMENTS

GF	Temperature	Humidity	Pressure	Pa
02	TT-GF-02	RHT-GF-02	DPT-GF-02	50
05	TT-GF-05	RHT-GF-05	DPT-GF-05	50
04	TT-GF-04	RHT-GF-04	DPT-GF-04	50
08	TT-GF-08	RHT-GF-08	DPT-GF-08	50
06	TT-GF-06	RHT-GF-06	DPT-GF-06	50
07	TT-GF-07	RHT-GF-07	DPT-GF-07	50
09	TT-GF-09	RHT-GF-09	DPT-GF-09	25
03	TT-GF-03	RHT-GF-03	DPT-GF-03	25
01	TT-GF-01	RHT-GF-01	DPT-GF-01	25

MAIN DUCT / PLANT MONITORING

TT-SA-01 / TT-RA-01: main duct temperature

RHT-RA-01: main return humidity

FIT-SA-01 / FIT-RA-01: common airflow

Each PAU filter: PDT remote + PDI local

DPT-SF-A/B/C: PAU fan pressure proof

VS: PAU and ventilation-fan vibration

EM: feeders >7.5 kW, quantity to confirm

Signals report to HPCP-01 dedicated PLC and HMI-01. BMS / SCADA provides the specified monitoring, initial start and emergency-stop interfaces. Protocol, alarm limits, I/O module allocation and actuator fail positions remain pending.

DPT room references are outdoors, not the adjacent room. DPT-GF-06 is the corridor / building reference. Additional room DPTs and CAG humidity sensing are marked as added provisions in the register.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOM INSTRUMENTATION AND CONTROL LOCATIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-007

SCALE: 1:50 AT A1

REVISION

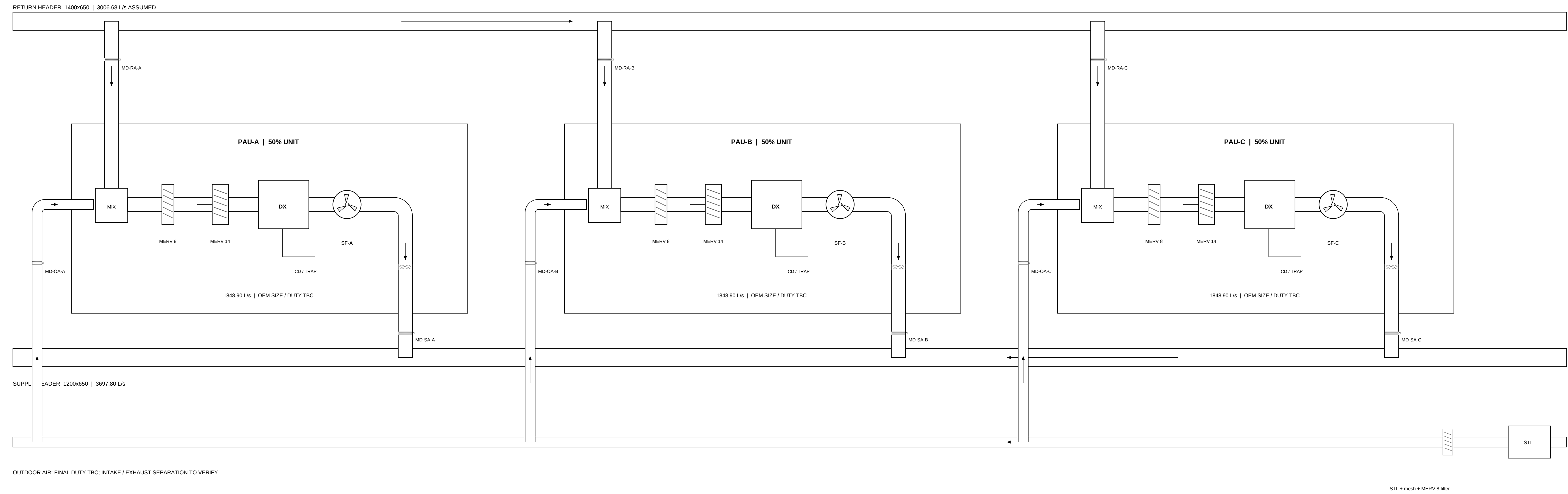
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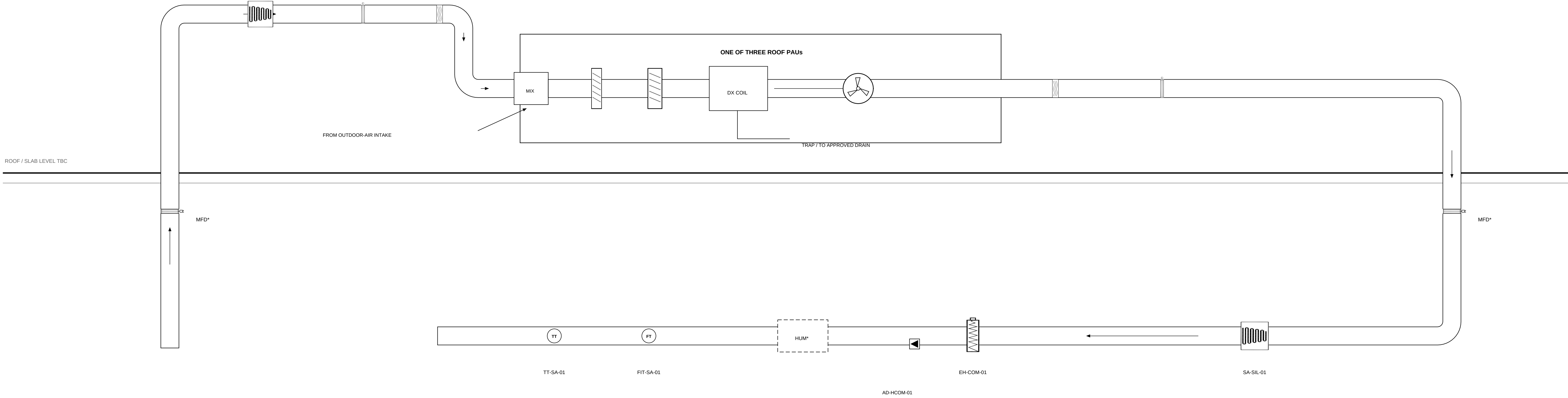
SHEET 7 OF 18

ADMINISTRATION BUILDING - PAU BANK AND INDOOR RISER SECTION

A ROOF CONNECTION ARRANGEMENT - SCHEMATIC FOOTPRINTS



B. ONE PAU / RISERS / INDOOR COMMON SUPPLY SECTION



All unit footprints / levels are schematic. MFD* requires approved compartmentation. Common heater and conditional humidifier are indoors. Filter DP, fan pressure / vibration and heater proof instruments: D-012 / S-001.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PAU BANK AND INDOOR RISER SECTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-008

SCALE: NTS AT A1

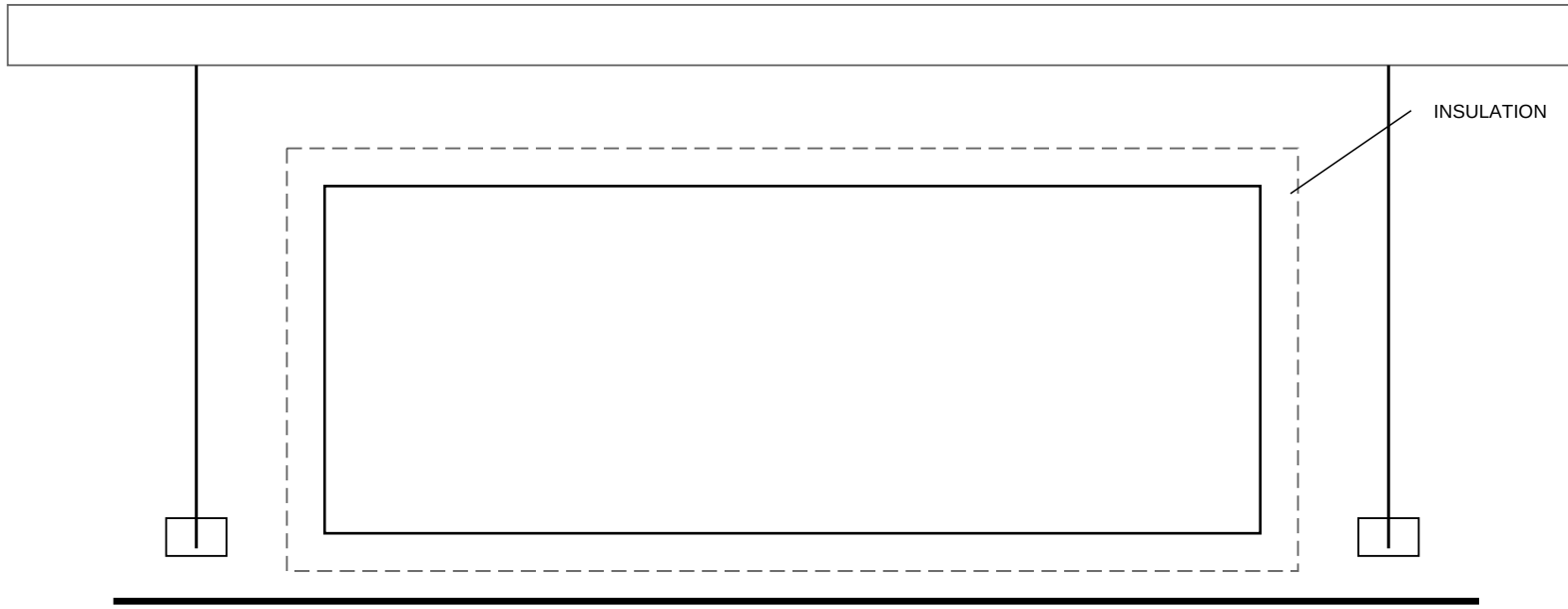
REVISION

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07 OCT 2026

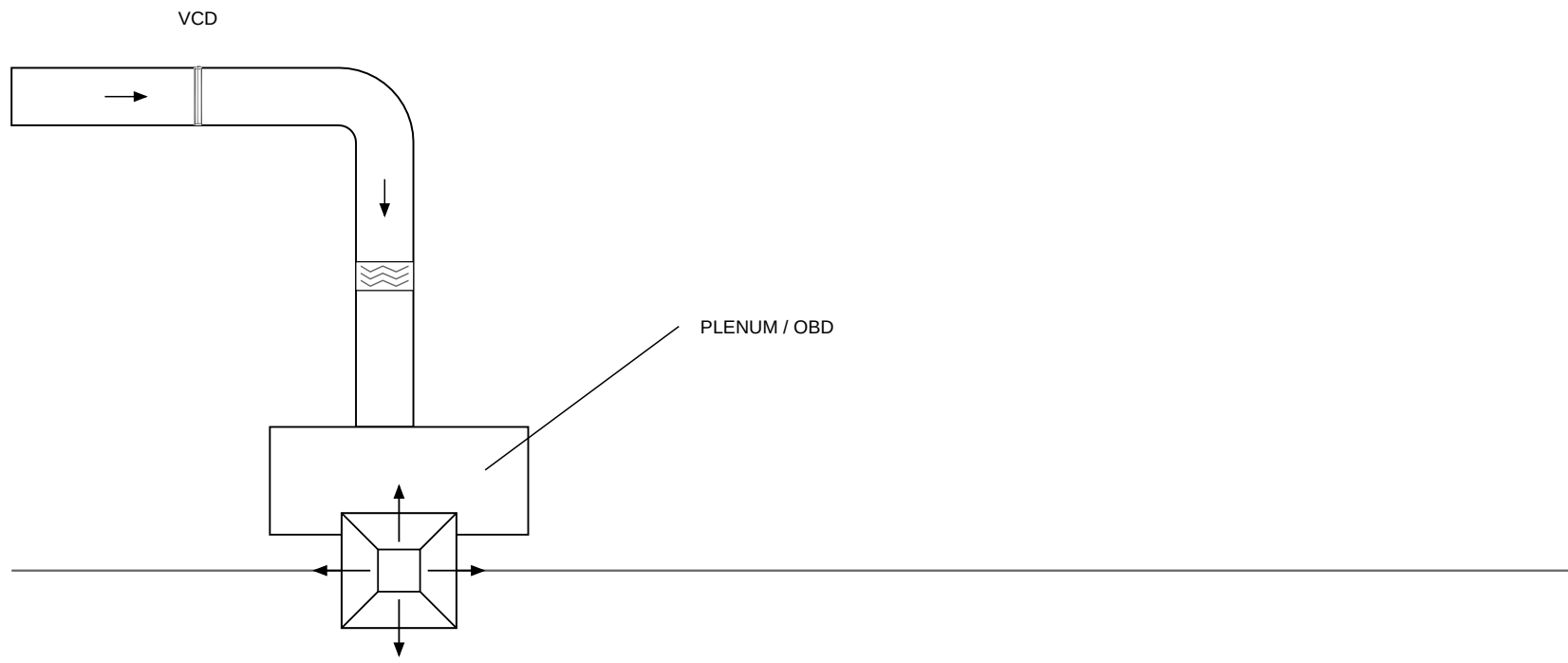
SHEET 8 OF 18

01 DUCT SUPPORT / INSULATION



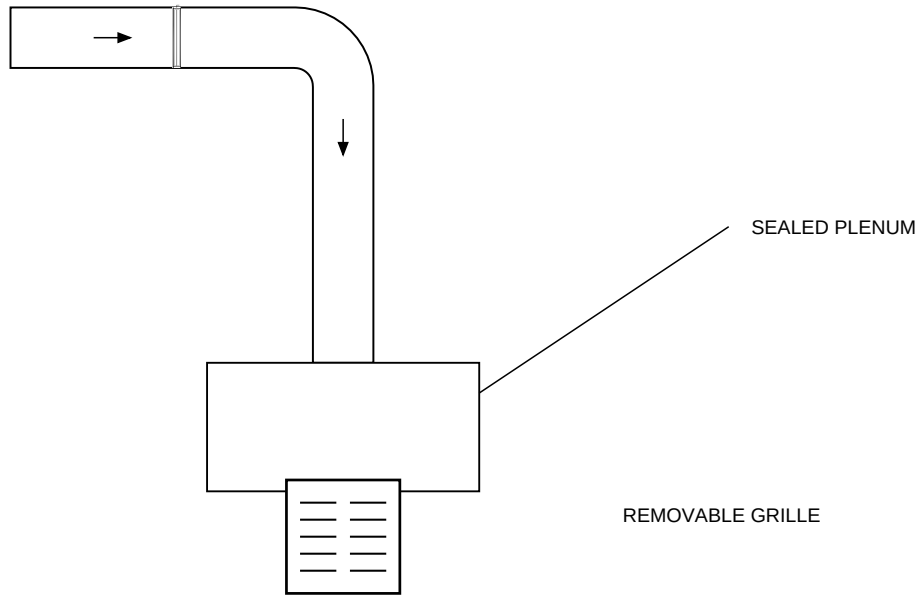
Anchor loads, support spacing, insulation and vapor barrier: approved fabrication / structural selection.

02 SUPPLY DIFFUSER CONNECTION



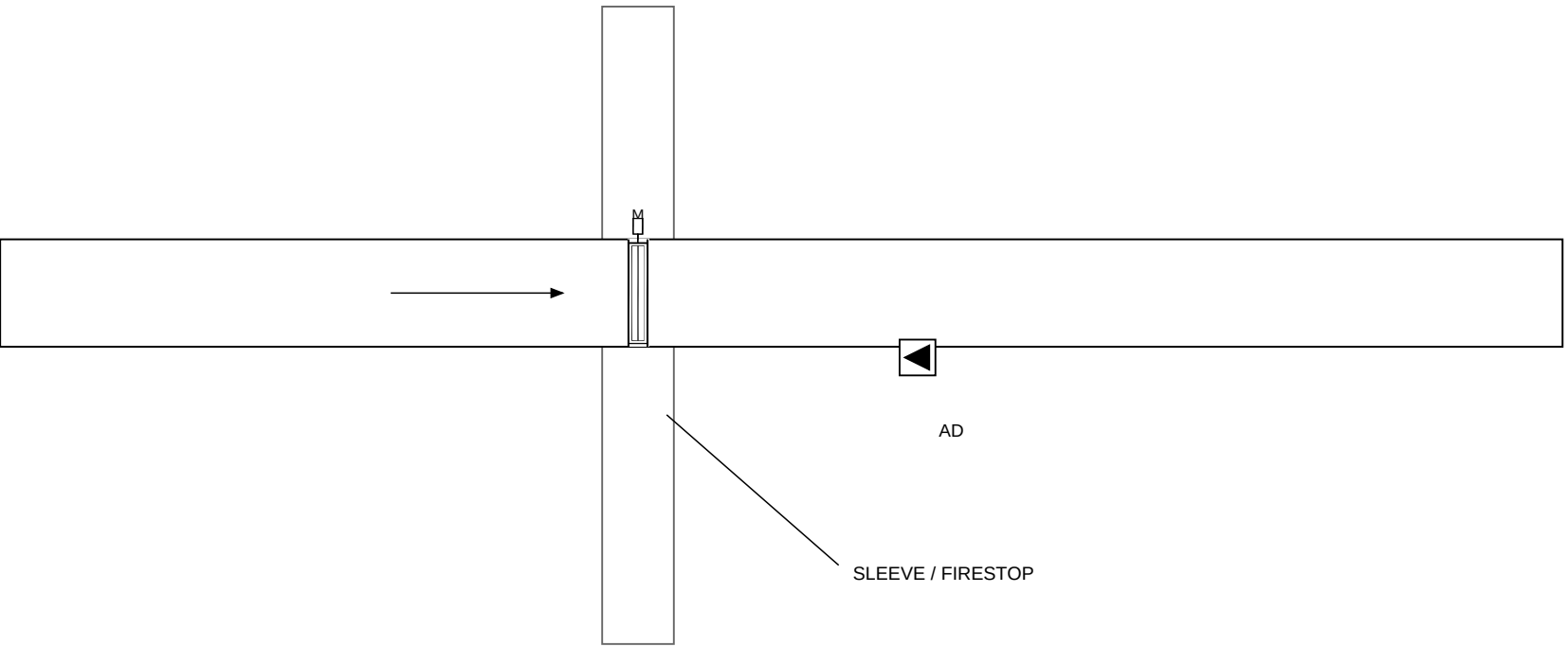
Neck <=2 m/s. Face, throw, noise and loss from selected diffuser catalogue; accessible balancing required.

03 RETURN / EXTRACT GRILLE



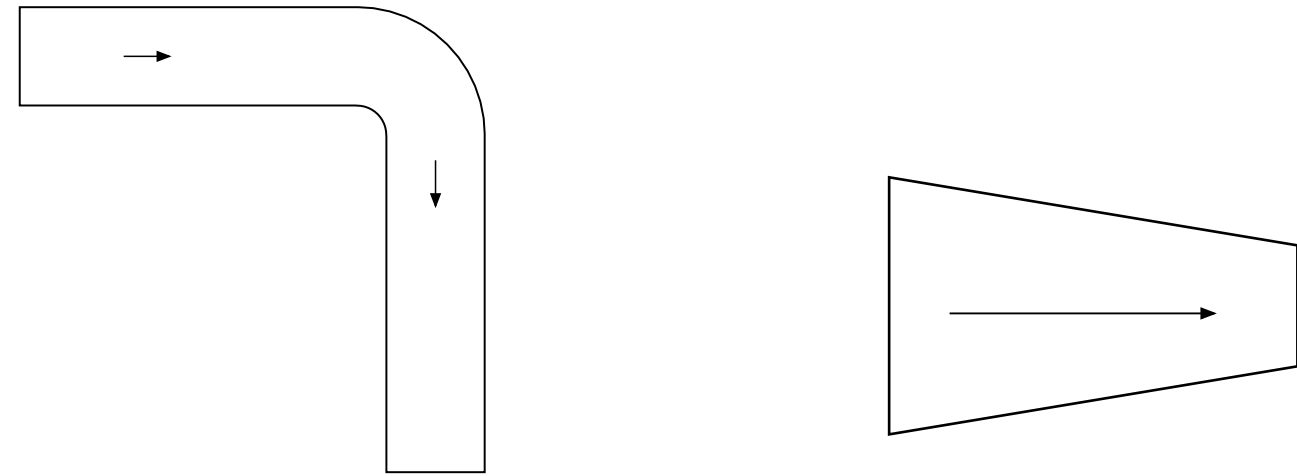
Return grilles connect to RA duct. Extract grilles connect to their dedicated outdoor duct; pickup height TBC.

04 RATED PENETRATION / MFD*



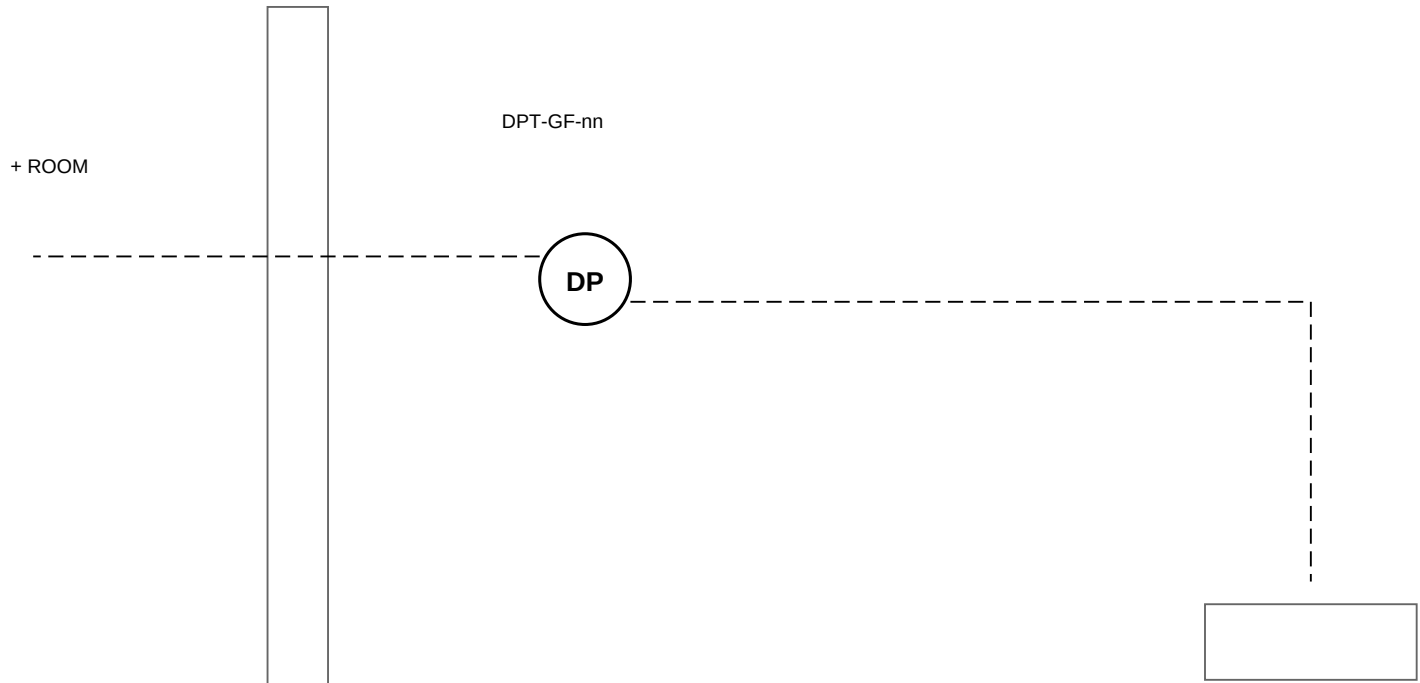
MFD* only at approved rated boundaries. Select tested sleeve / firestop assembly and allow actuator / inspection access.

05 BEND / TAPER TRANSITION



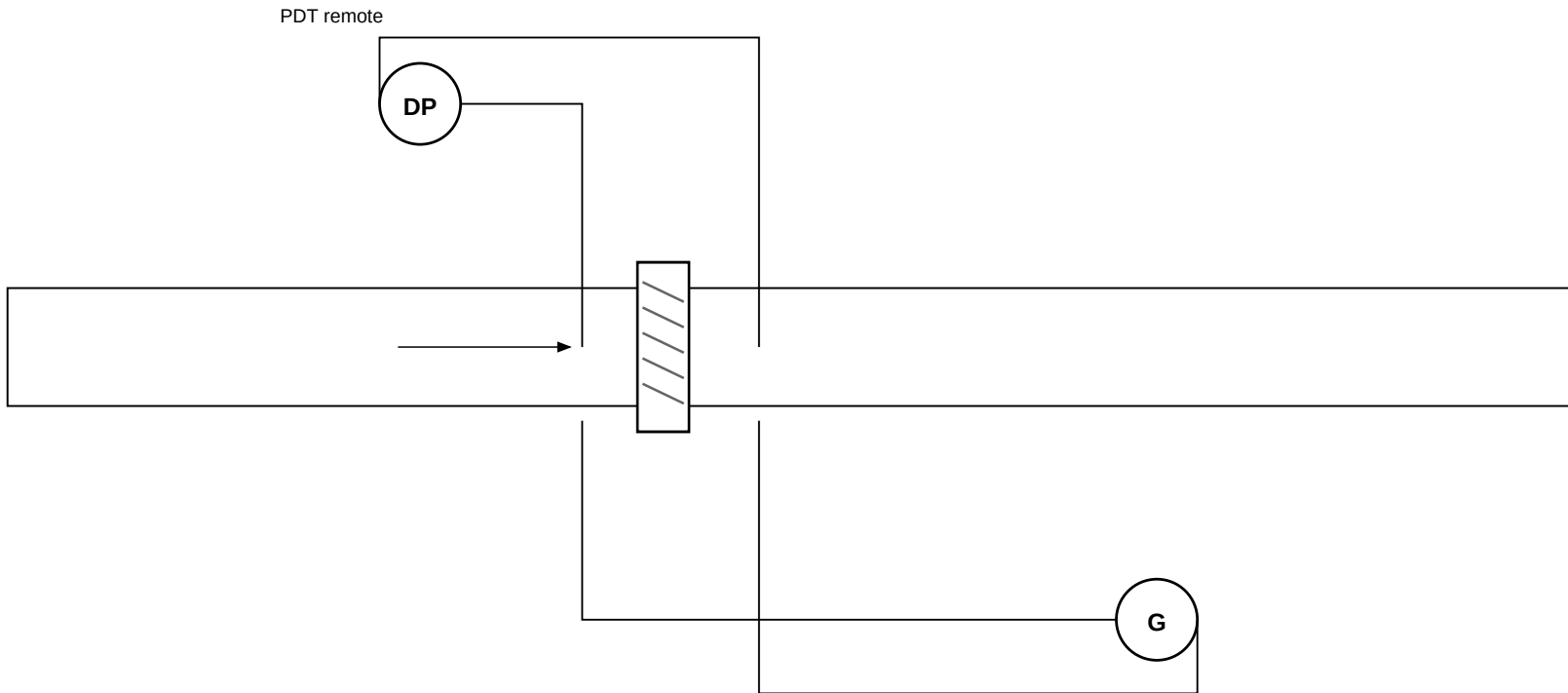
Radius bends or miter bends / turning vanes to selected fitting design. Final loss coefficients and development TBC.

06 ROOM PRESSURE SENSING



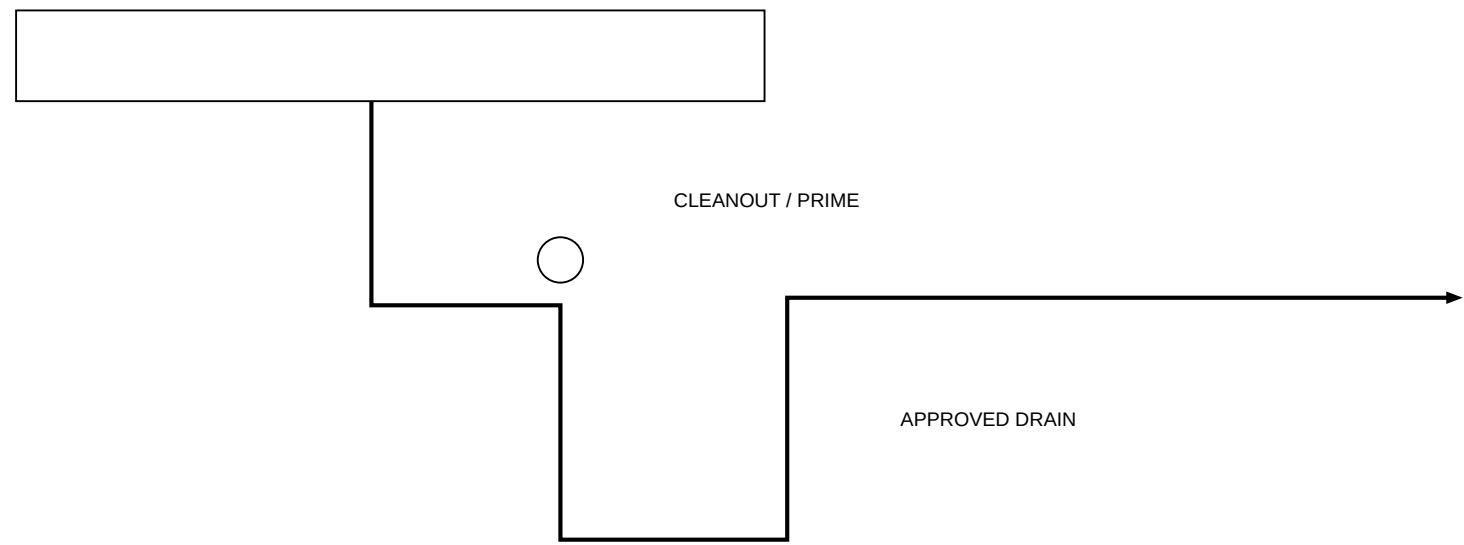
Weather-shielded outdoor static reference; route tubes to minimize wind / water effects. Range, alarm limits and mounting TBC.

07 FILTER DP MONITORING



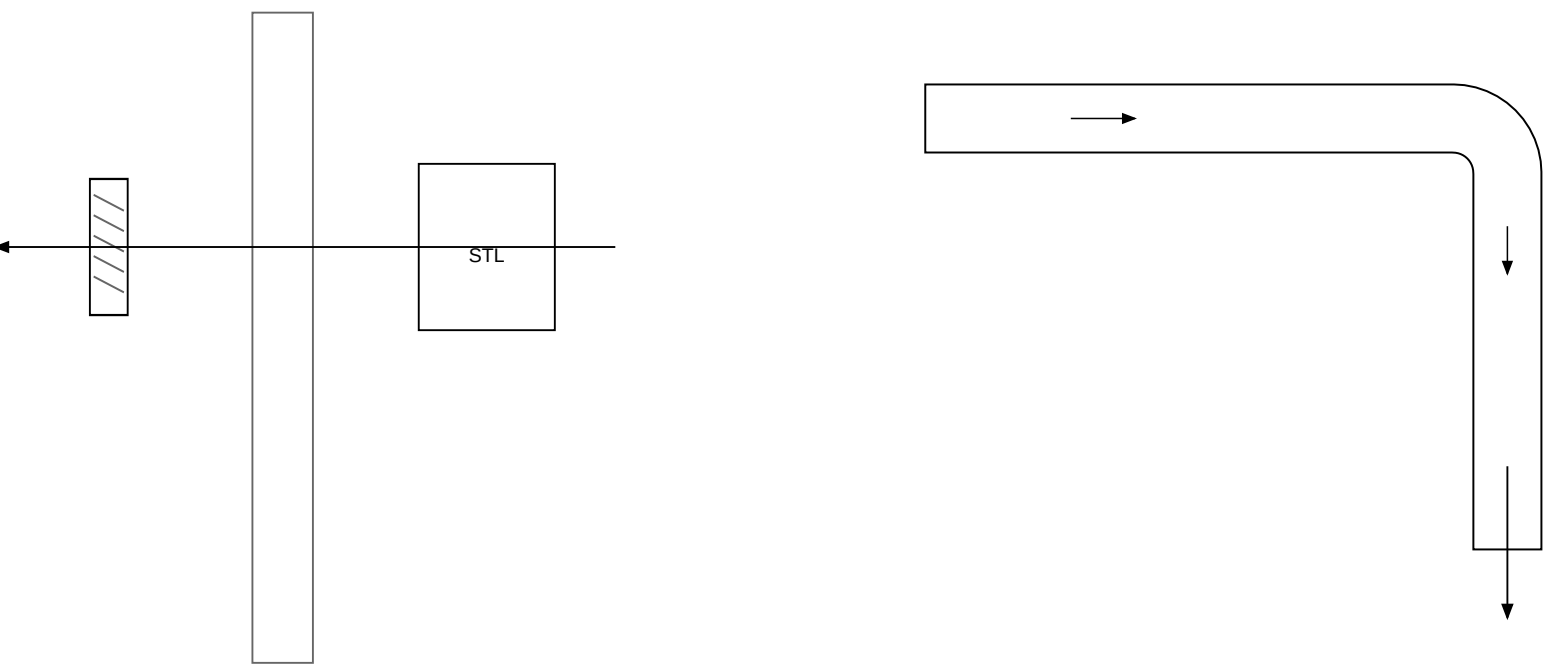
Each PAU filter bank has remote DP and local gauge. Upstream / downstream taps; dirty-filter threshold by OEM.

08 COIL DRAIN / TRAP



Trap seal / invert from actual unit casing pressure. Drain slope, disposal route, cleanout and water seal by OEM / plumbing.

09 INTAKE / EXHAUST OUTLET



STL face <=1 m/s; exhaust louver <=2.5 m/s. Final face areas, loss, materials and intake separation require selection.

Details are NTS engineering concepts. Duct colours / symbols match the plans; body dimensions are not manufacturer selections. No room pressure, fire rating or equipment clearance is certified by this drawing.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DUCT SYMBOLS AND INSTALLATION DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-009

SCALE: NTS AT A1

REVISION

09

07 OCT 2026

SHEET 9 OF 18

ADMINISTRATION BUILDING - DUCT AND TERMINAL SCHEDULES

MAIN AND ROOM BRANCH SECTIONS

Section	Size mm	L/s	m/s	Lh m*	Pa/m
SA-M01	1200x650	3697.80	4.74	2.40	0.252
SA-M02	1100x575	3033.00	4.80	3.40	0.294
SA-M03	950x500	2255.30	4.75	5.60	0.343
SA-M04	850x425	1806.10	5.00	3.50	0.449
SA-M05	650x300	921.50	4.73	4.00	0.599
SA-M06	500x250	503.30	4.03	2.20	0.575
SA-M07	450x250	457.10	4.06	2.30	0.611
SA-M08	400x200	288.90	3.61	2.30	0.619
SA-M09	300x200	141.30	2.35	0.70	0.325
RA-M01	1400x650	3006.68	3.30	2.60	0.122
RA-M02	1200x600	2401.22	3.34	4.20	0.140
RA-M03	850x550	1682.85	3.60	4.45	0.201
RA-M04	800x450	1292.99	3.59	3.40	0.239
RA-M05	650x300	467.73	2.40	3.80	0.173
RA-M06	450x225	108.86	1.08	4.05	0.060
SA-B02	500x350	664.80	3.80	3.00	0.399
RA-B02	600x400	605.46	2.52	2.50	0.157
SA-B04	450x300	449.20	3.33	3.40	0.369
RA-B04	500x350	389.86	2.23	2.55	0.151
SA-B08	600x400	884.60	3.69	4.50	0.313
RA-B08	700x450	825.26	2.62	2.85	0.143
SA-B05	600x350	777.70	3.70	12.20	0.349
RA-B05	650x400	718.36	2.76	11.40	0.178
SA-B06	450x300	418.20	3.10	2.10	0.324
RA-B06	550x350	358.86	1.86	6.05	0.104
SA-B09	200x150	46.20	1.54	2.40	0.231
SA-B07	300x250	168.20	2.24	7.10	0.254
RA-B07	350x250	108.86	1.24	6.50	0.081
SA-B03	300x200	147.60	2.46	2.40	0.351
SA-B01	300x200	141.30	2.35	7.50	0.325
EA-B09	250x200	64.89	1.30	6.70	0.123
EA-B03	300x200	147.60	2.46	1.90	0.351
EA-B01	TBC	TBC	TBC	TBC	TBC

* Lh is the source orthogonal horizontal route allowance. Final straight lengths, selected elbow development, risers / drops and fitting losses require coordination. Vertical lengths and BOD are unverified.

Sizes are clear internal dimensions. Catalogue free area, diffuser throw / noise, coil / filter / louver / silencer losses, fitting coefficients and OEM fan curves are pending. Confirmed fire stops the entire HVAC.

TERMINAL GROUPS

GF	Type	Qty	Each L/s	Neck mm	Face mm
02	SD	4	166.20	D350	600x600
02	RG	4	151.37	D300	500x500
05	SD	5	155.54	D325	600x600
05	RG	5	143.67	D300	500x500
04	SD	3	149.73	D325	600x600
04	RG	3	129.95	D300	500x500
08	SD	6	147.43	D325	600x600
08	RG	6	137.54	D300	500x500
06	SD	3	139.40	D300	600x600
06	RG	4	89.72	D300	500x500
07	SD	1	168.20	D350	600x600
07	RG	1	108.86	D300	500x500
09	SD	2	23.10	D150	300x300
09	EG	2	32.44	D150	300x300
03	SD	1	147.60	D325	600x600
03	EG	1	147.60	D300	500x500
01	SD	1	141.30	D300	600x600
01	EG	1	TBC	TBC	500x500

SYSTEM / DESIGN BASIS

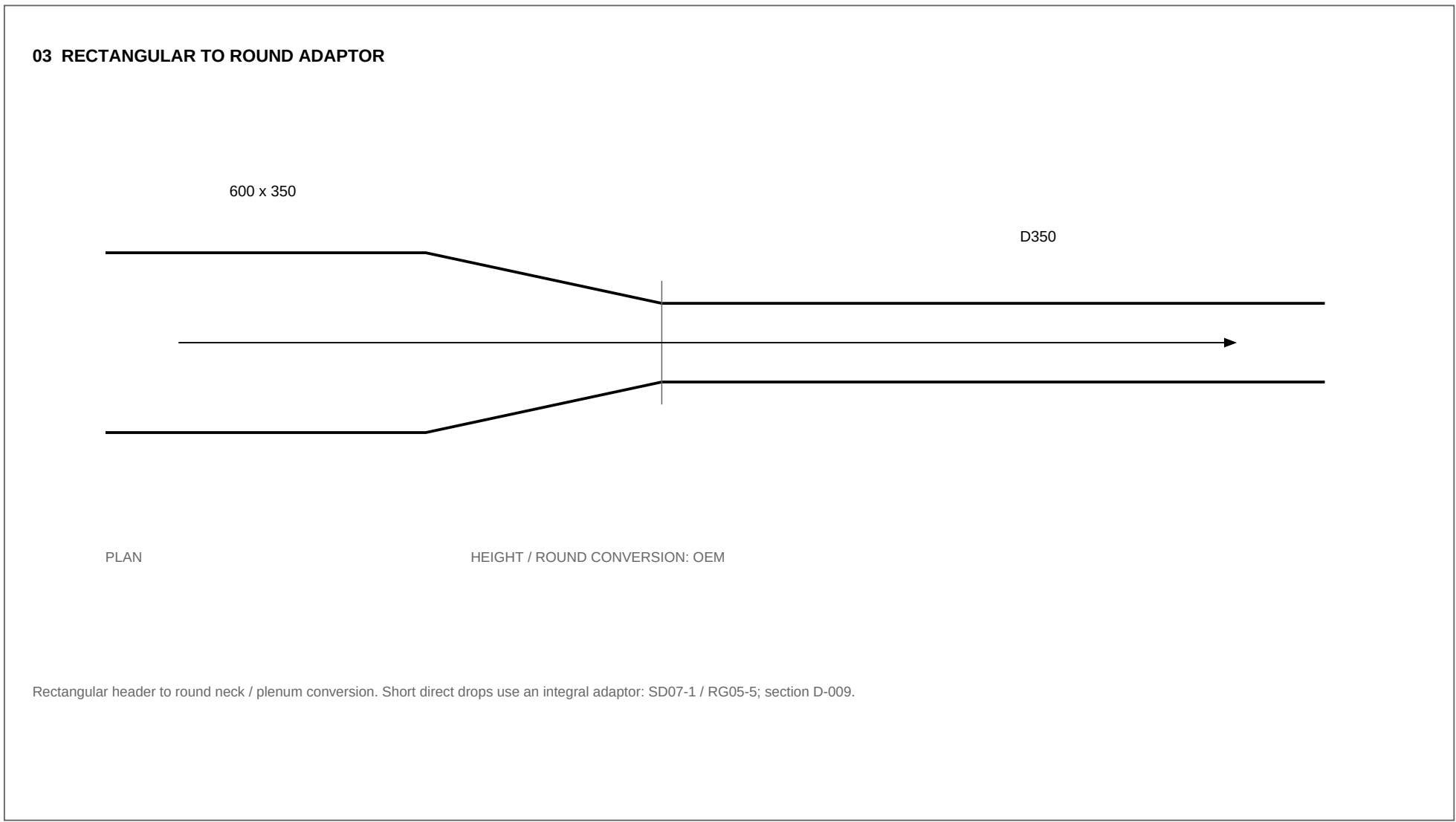
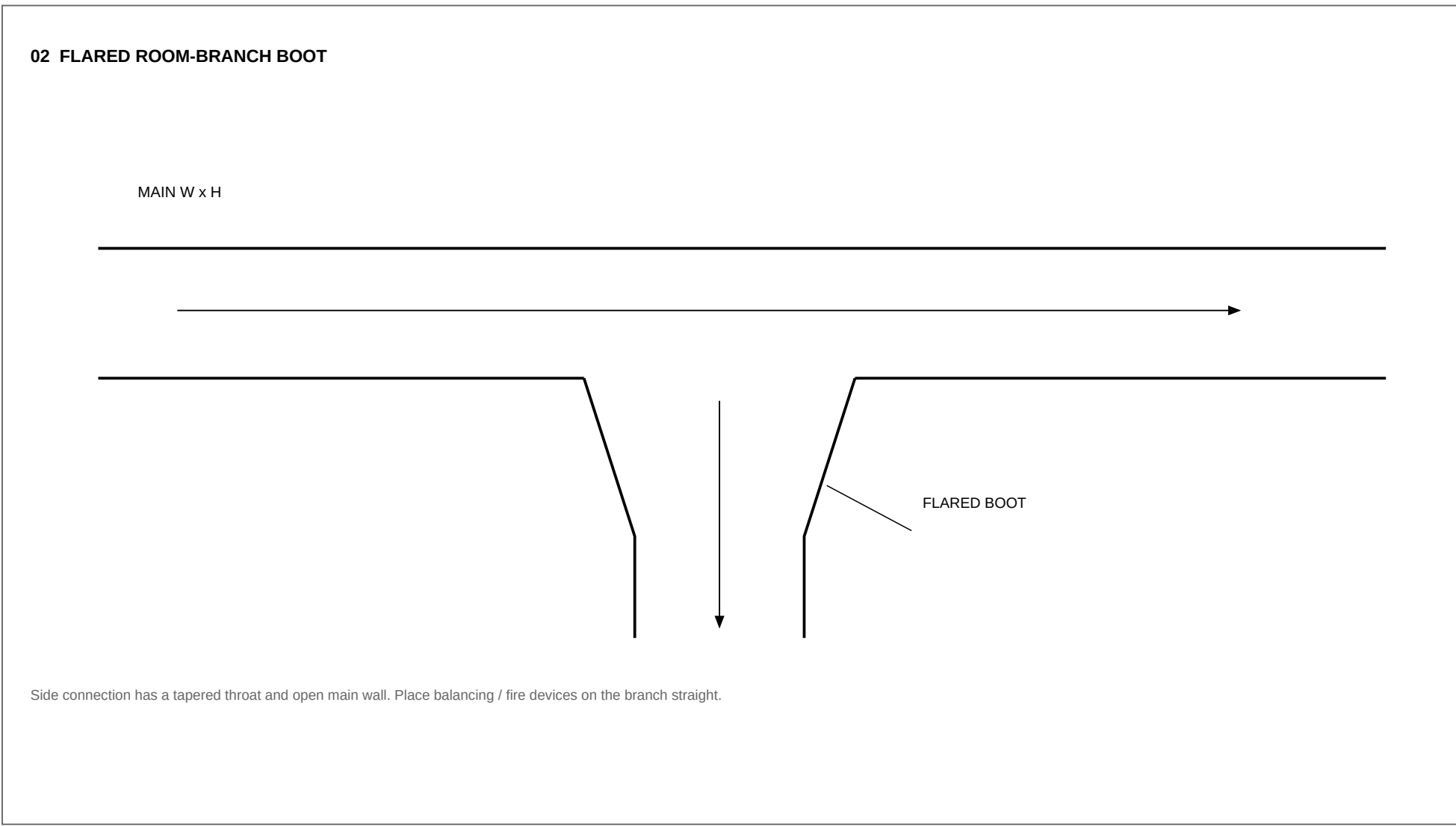
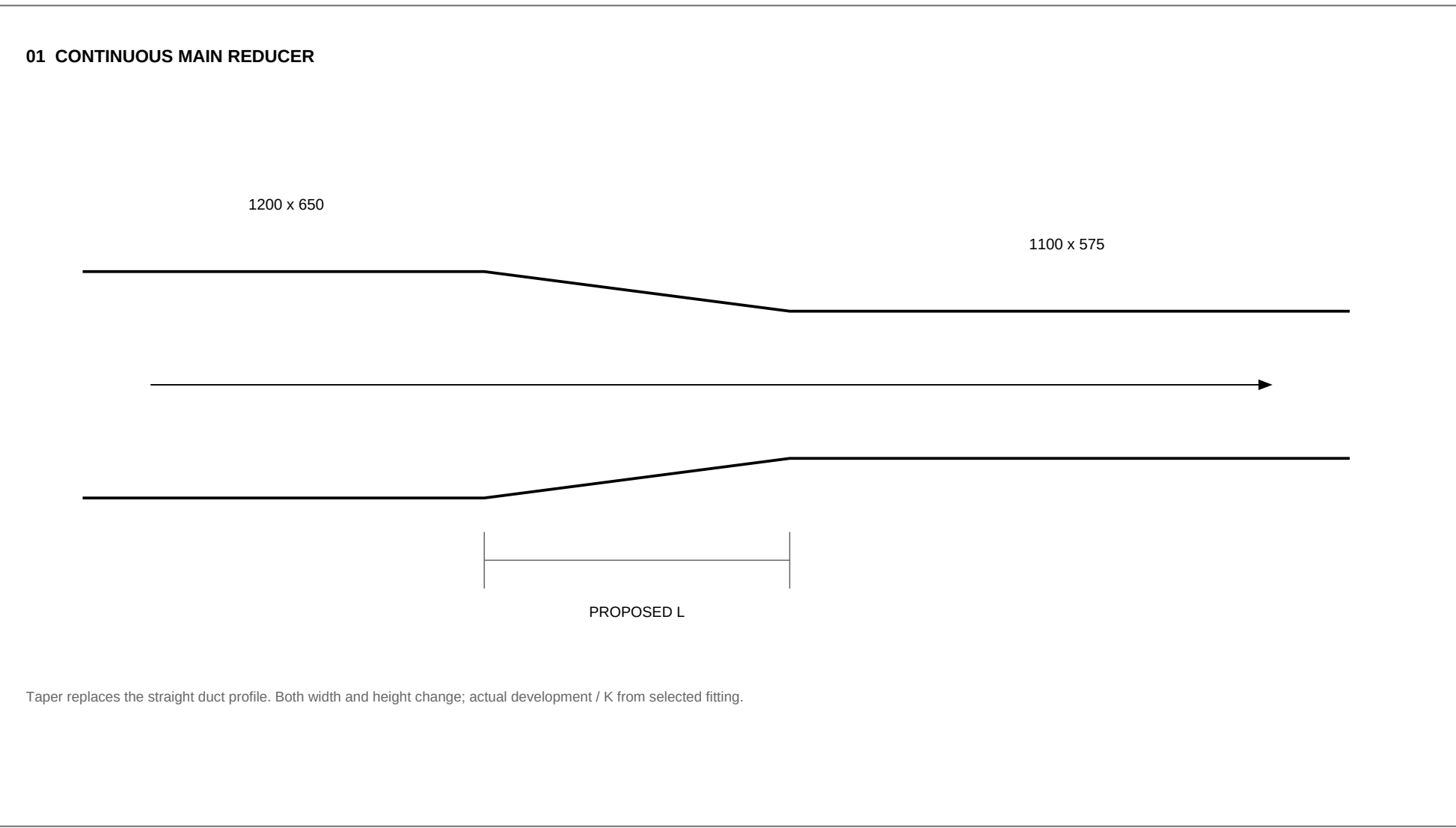
Total supply: 3697.80 L/s. Assumed return: 3006.68 L/s.

Six return rooms: 59.34 L/s net outward allowance each, based on the accepted 0.01 m2 illustrative equivalent area.

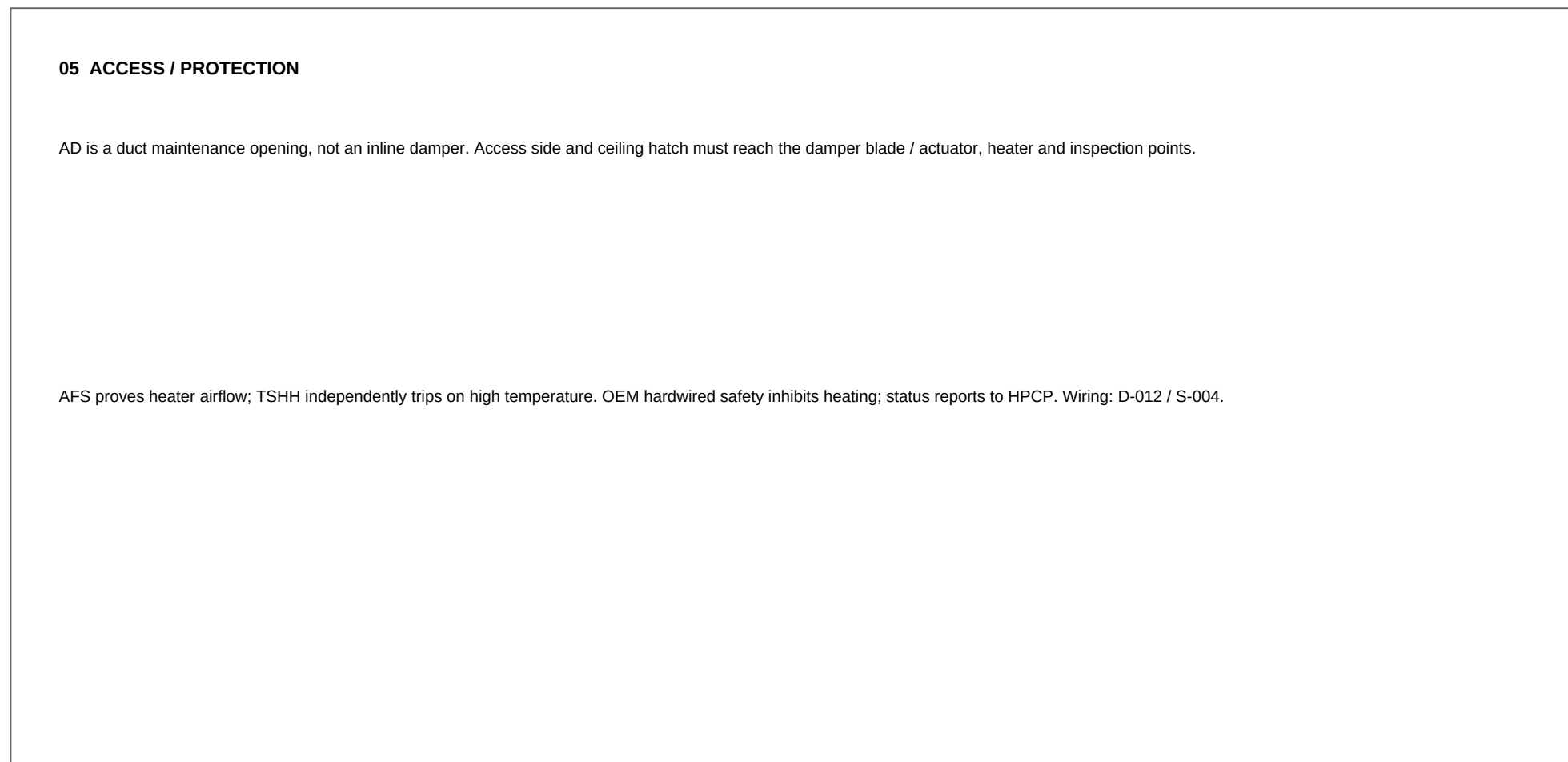
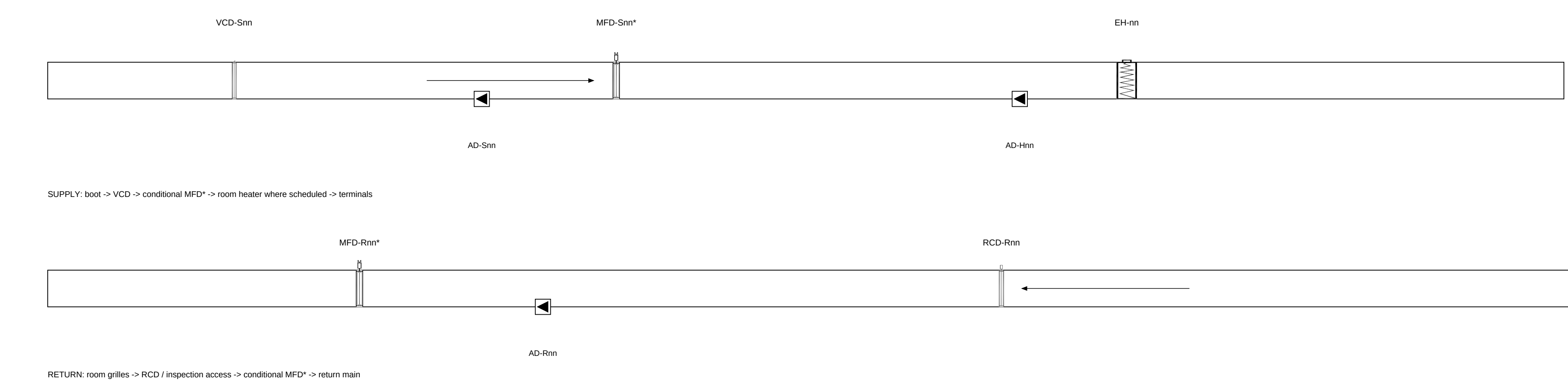
Final room pressure, IAQ outdoor-air duty and revised HAP loads remain unverified. Current 420.70 L/s OA is below the 691.12 L/s assumed-balance makeup requirement.

All 114 duct sections, 100 fitting entries, 53 terminals, 176 components and 85 instruments are in the editable registers.

Rev09: engineering presentation and complete instrument referencing added. Rev08 duct connections and device positions retained; see D-013 to D-018.



04 TYPICAL BRANCH STATIONS - FLOW DIRECTION / ACCESS



MAIN TAPER SCHEDULE - PROPOSED GEOMETRY

Fitting	Inlet mm	Outlet mm	L mm*
SA-RED-01	1200x650	1100x575	350
SA-RED-02	1100x575	950x500	375
SA-RED-03	950x500	850x425	350
SA-RED-04	850x425	650x300	500
SA-RED-05	650x300	500x250	375
SA-RED-06	500x250	450x250	350
SA-RED-07	450x250	400x200	350
SA-RED-08	400x200	300x200	350
RA-RED-01	1200x600	1400x650	500
RA-RED-02	850x550	1200x600	875
RA-RED-03	800x450	850x550	350
RA-RED-04	650x300	800x450	375
RA-RED-05	450x225	650x300	500

* Taper lengths are explicit proposed coordination geometry, not approved fabrication lengths. The transition register includes main tapers, 15 branch boots and terminal adaptors; selected losses / height details remain pending.

ACCESS-DOOR TAGS / BRANCH MAP

GF	Supply access	Return access	Heater access
02	AD-S02	AD-R02	-
05	AD-S05	AD-R05	AD-H05
04	AD-S04	AD-R04	AD-H04
08	AD-S08	AD-R08	-
06	AD-S06	AD-R06	AD-H06
07	AD-S07	AD-R07	AD-H07
09	AD-S09	-	-
03	AD-S03	-	AD-H03
01	AD-S01	-	-
COM	AD-HCOM-01	-	Common heater

All 21 AD tags and device coordinates are registered. Door clear opening, sealing / insulation, access hatch and OEM working envelope remain to be selected.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

TAPERED FITTINGS AND MAINTENANCE ACCESS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-011

SCALE: NTS AT A1

REVISION

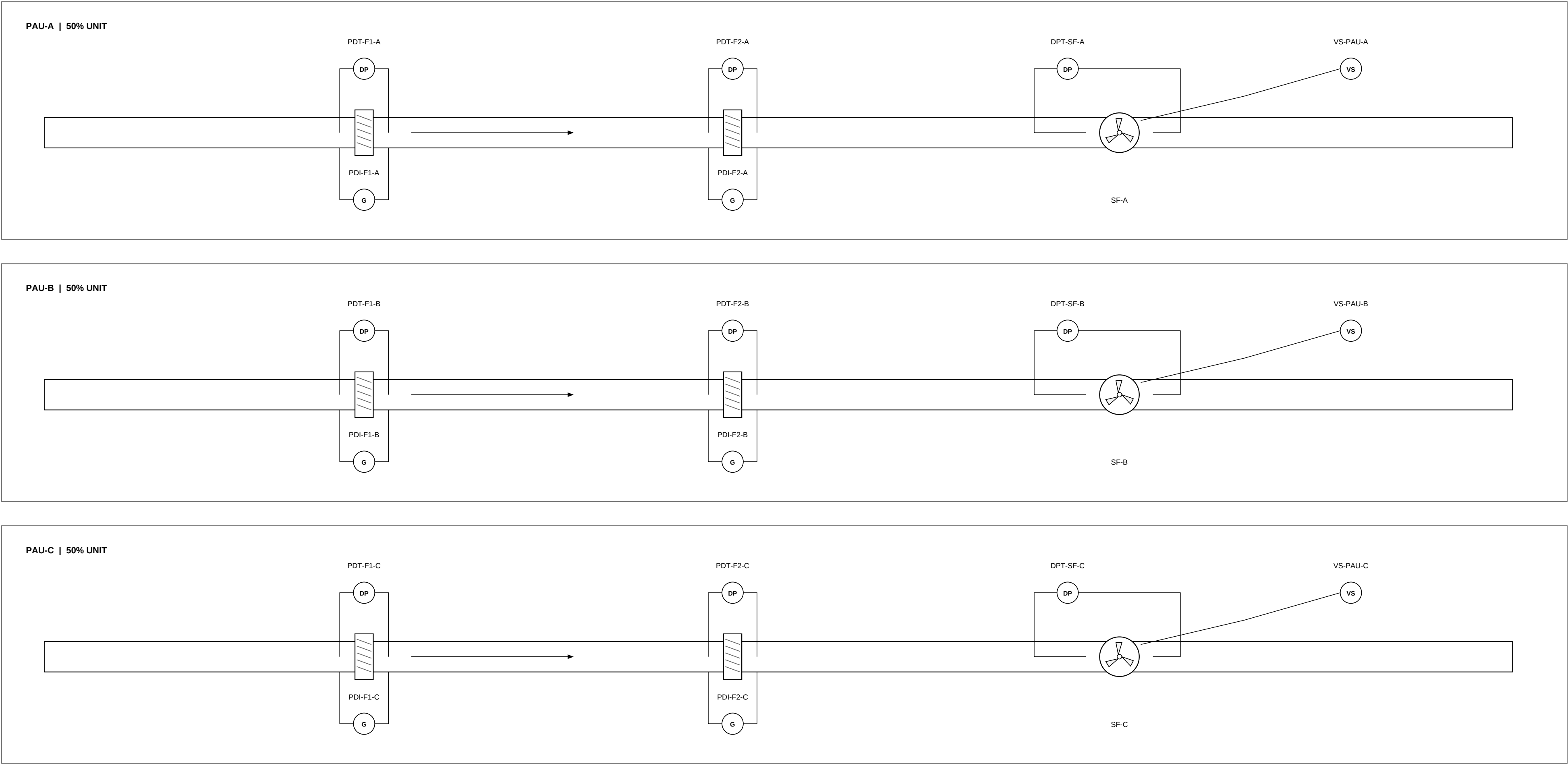
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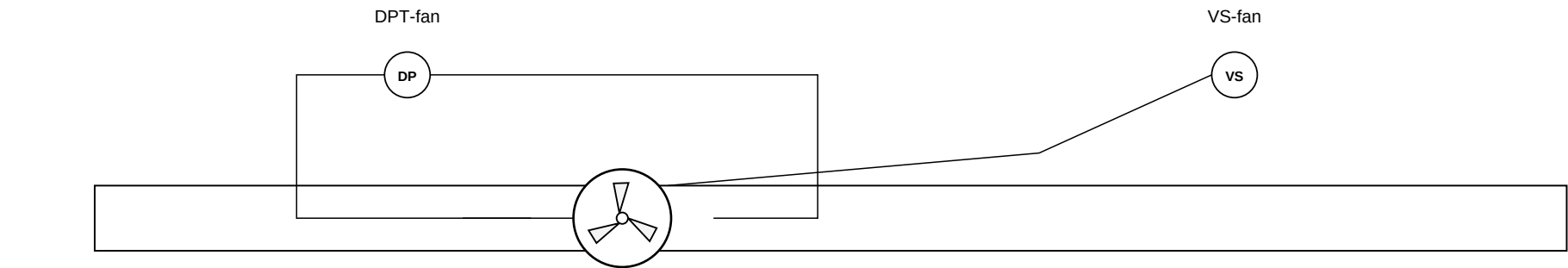
SHEET 11 OF 18

ADMINISTRATION BUILDING - PLANT INSTRUMENTATION AND HEATER PROTECTION

A EACH PAU: FILTER DIFFERENTIAL PRESSURE / FAN PROOF / VIBRATION



C EXTRACT / BASEMENT FAN PROOF

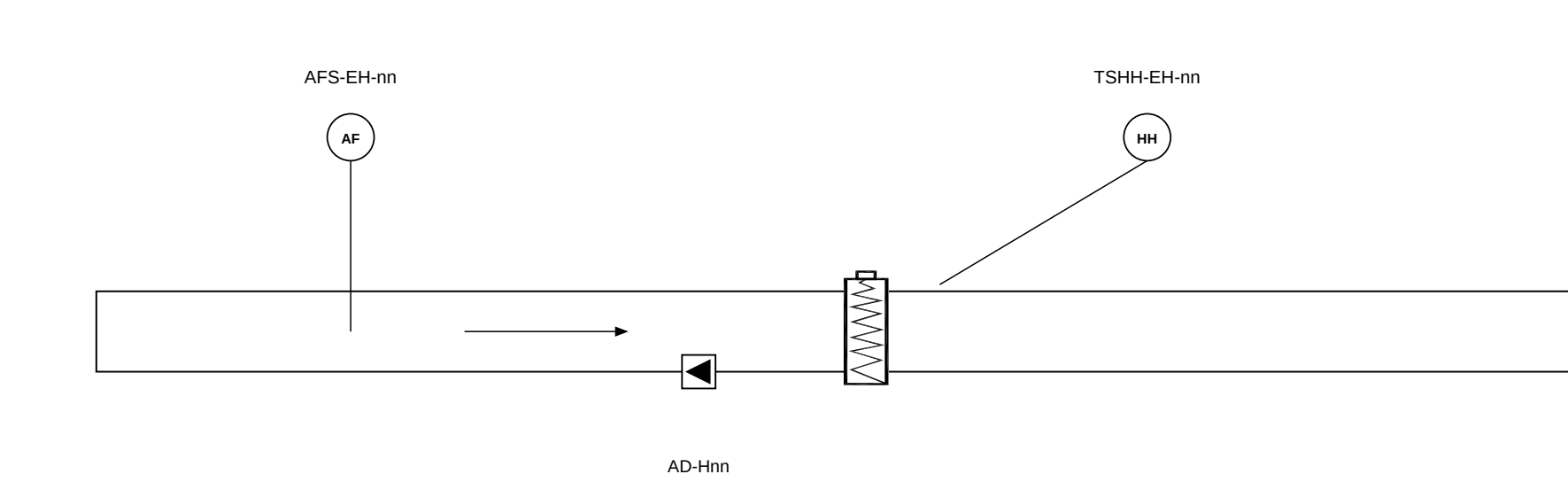


Tag suffixes: EF-03-01/02; EF-09-01/02; EF-01-01. Each has DPT- and VS- tags. Basement: SF-B1-01/02 and EF-B1-01/02 use the same convention.

Fan start / run / fault / ready / local-auto / E-stop and damper proofs are listed in I/O. CAG duty / quantity and basement duties remain pending.

Instrument glyphs are functional symbols, not equipment dimensions. AFS additions are explicit proposed OEM protection / monitoring provisions. Full tags: Excel / CSV registers. Room TT / RHT / DPT locations: D-007.

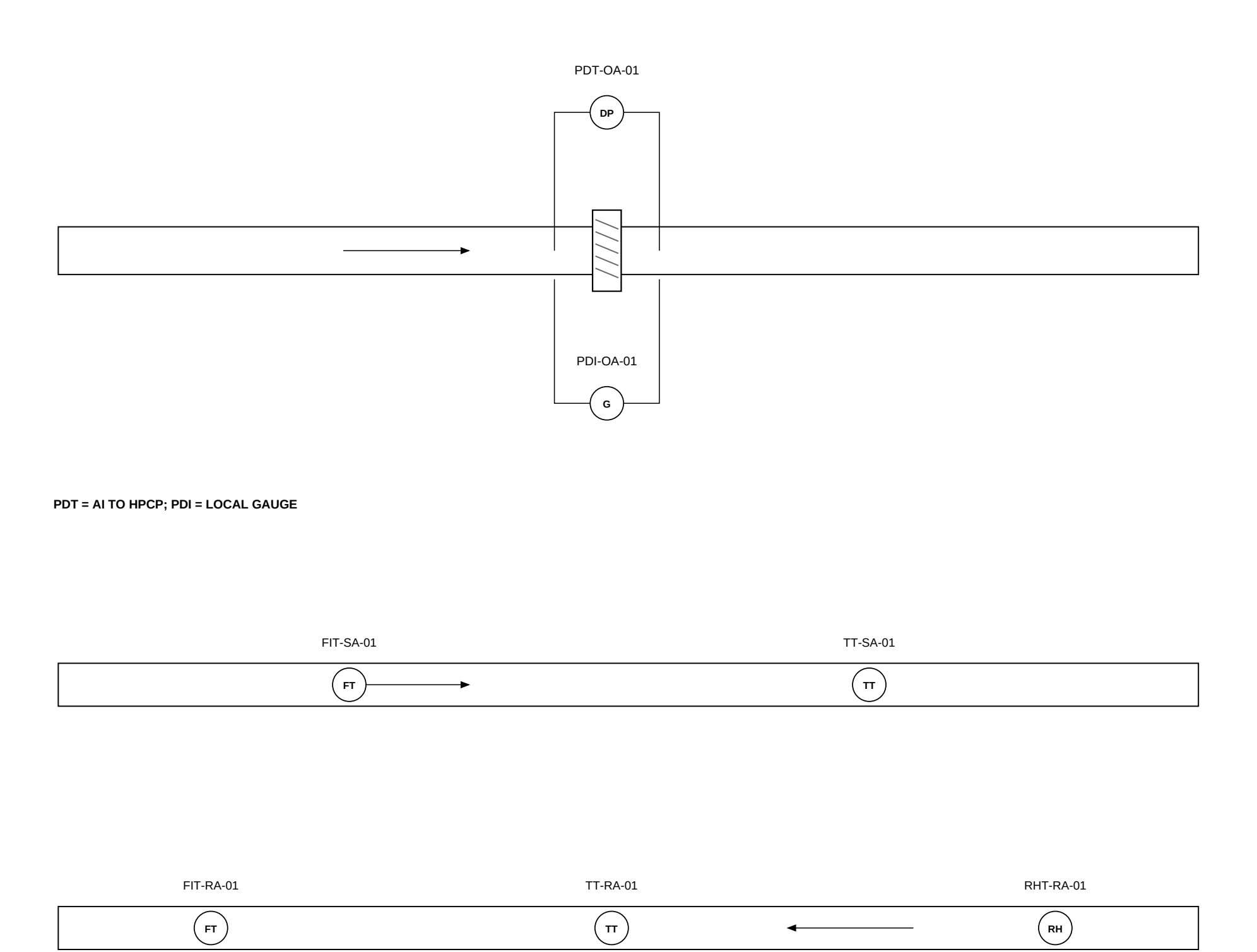
D HEATER AIRFLOW / HIGH-LIMIT SAFETY



nn = 04, 05, 06, 07, 03, COM-01. Six AFS DI points are retained from Rev07. Each heater retains independent TSHH; loss of proof / high limit switches heating off.

AFS proof technology, setpoint and OEM hardwired wiring to be selected. PLC demand cannot override local safety.

B OUTDOOR-AIR FILTER / COMMON DUCTS



Filter taps measure pressure across each filter, not between rooms. Fan DP proves the operating fan path; VS monitors fan / unit vibration. Alarm thresholds and OEM signal type TBC.

E PRELIMINARY HPCP I/O

Type	Points	Spare >=20%	Channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM: 3 interfaces; separate from wired channel totals. Signals report to dedicated HPCP-01 PLC / HMI-01 and BMS / SCADA. Final protocol and allocation TBC.

CONFIRMED FIRE: stop every HVAC fan and heater. No automatic fire extraction / post-fire purge. Reset and damper fire positions follow the approved cause-and-effect.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PLANT INSTRUMENTATION AND HEATER PROTECTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-012

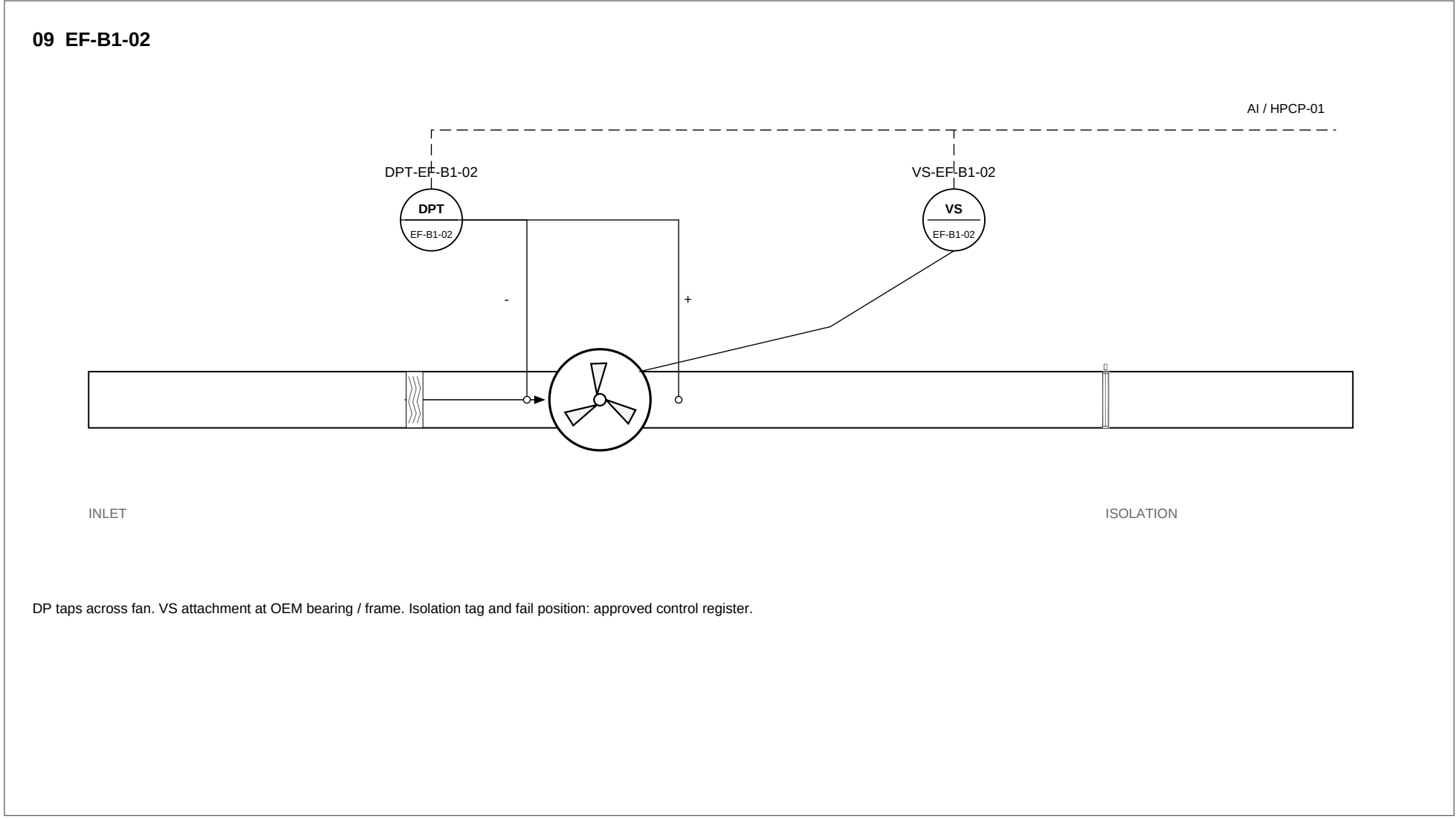
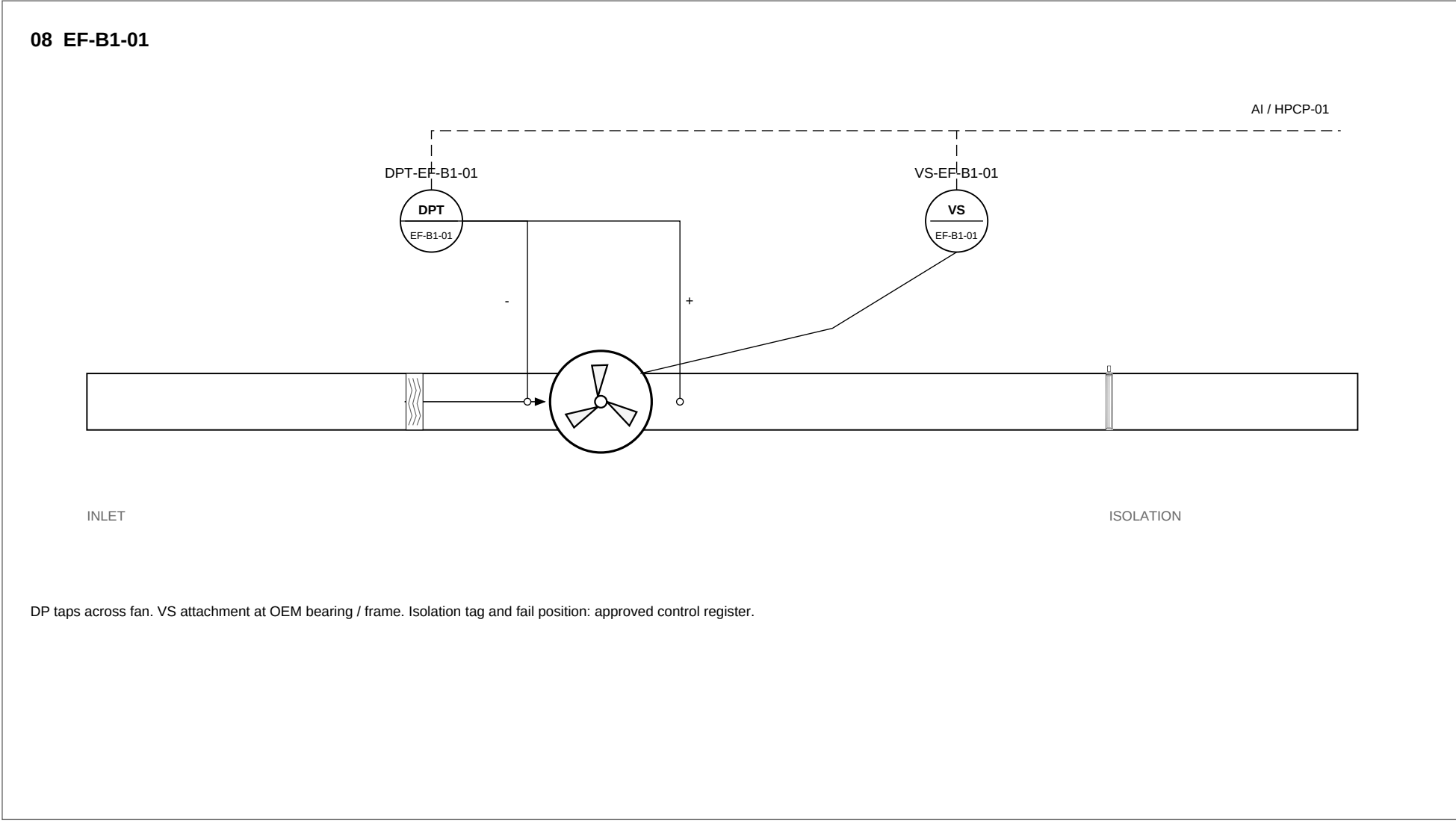
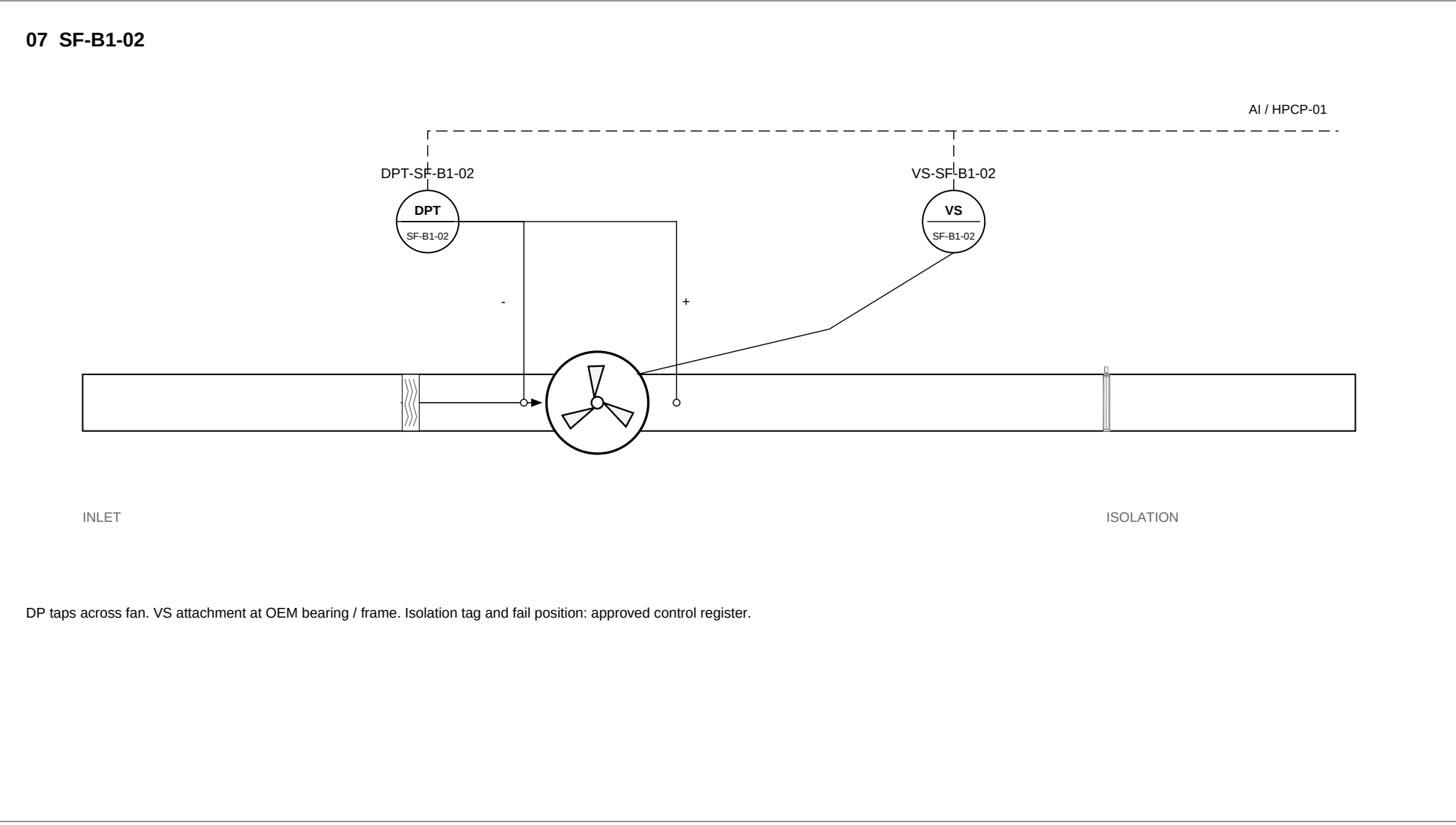
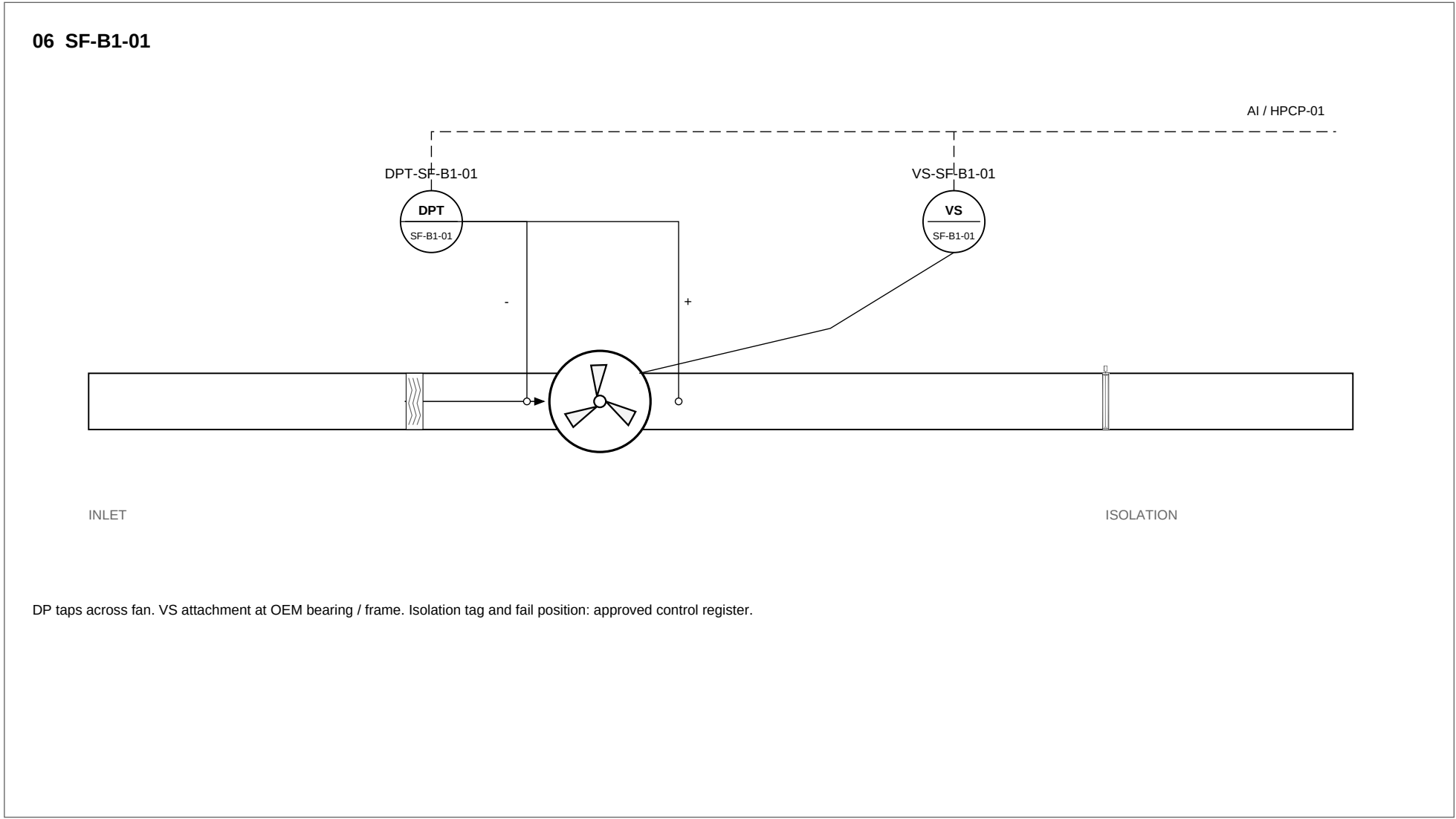
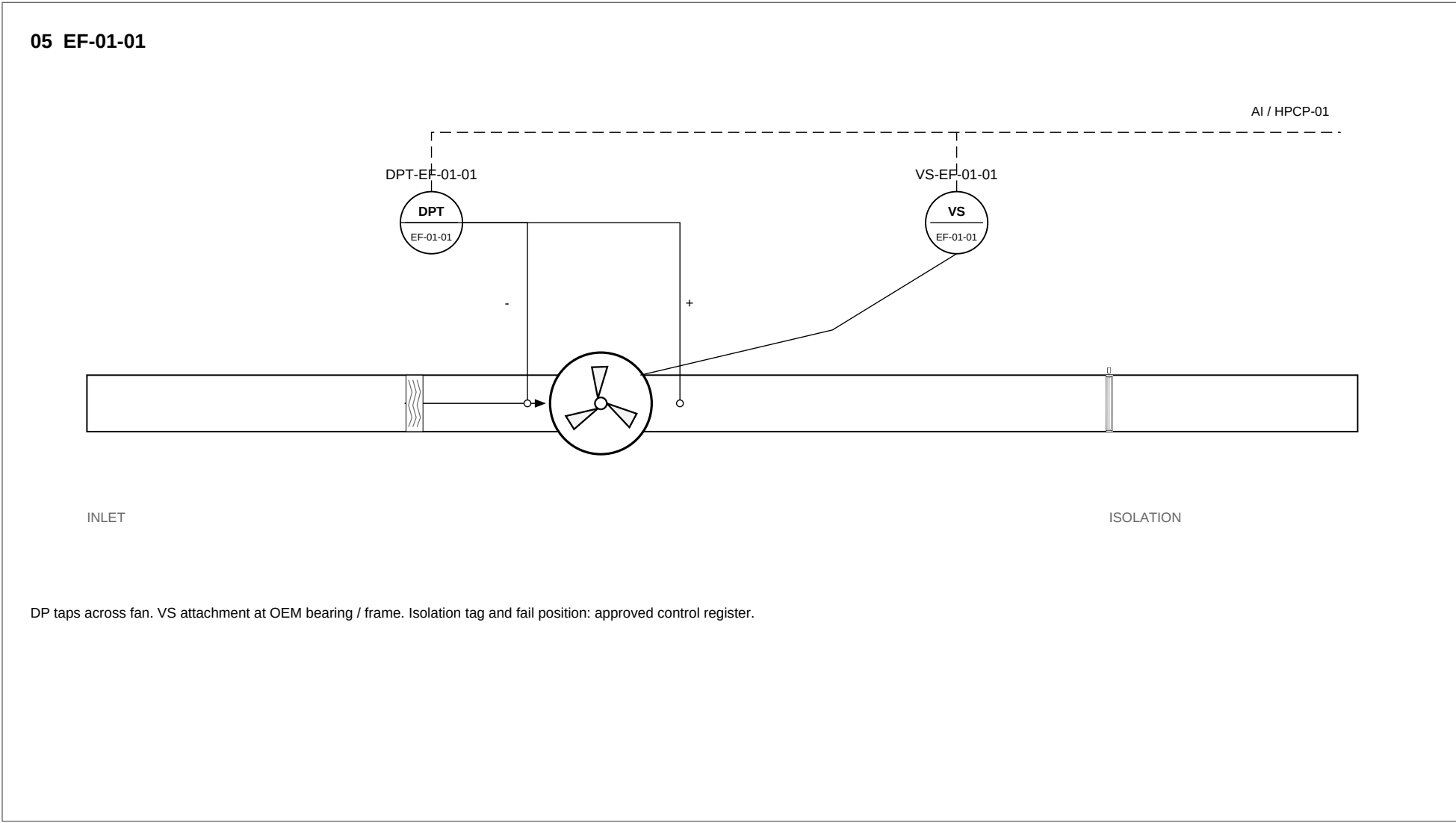
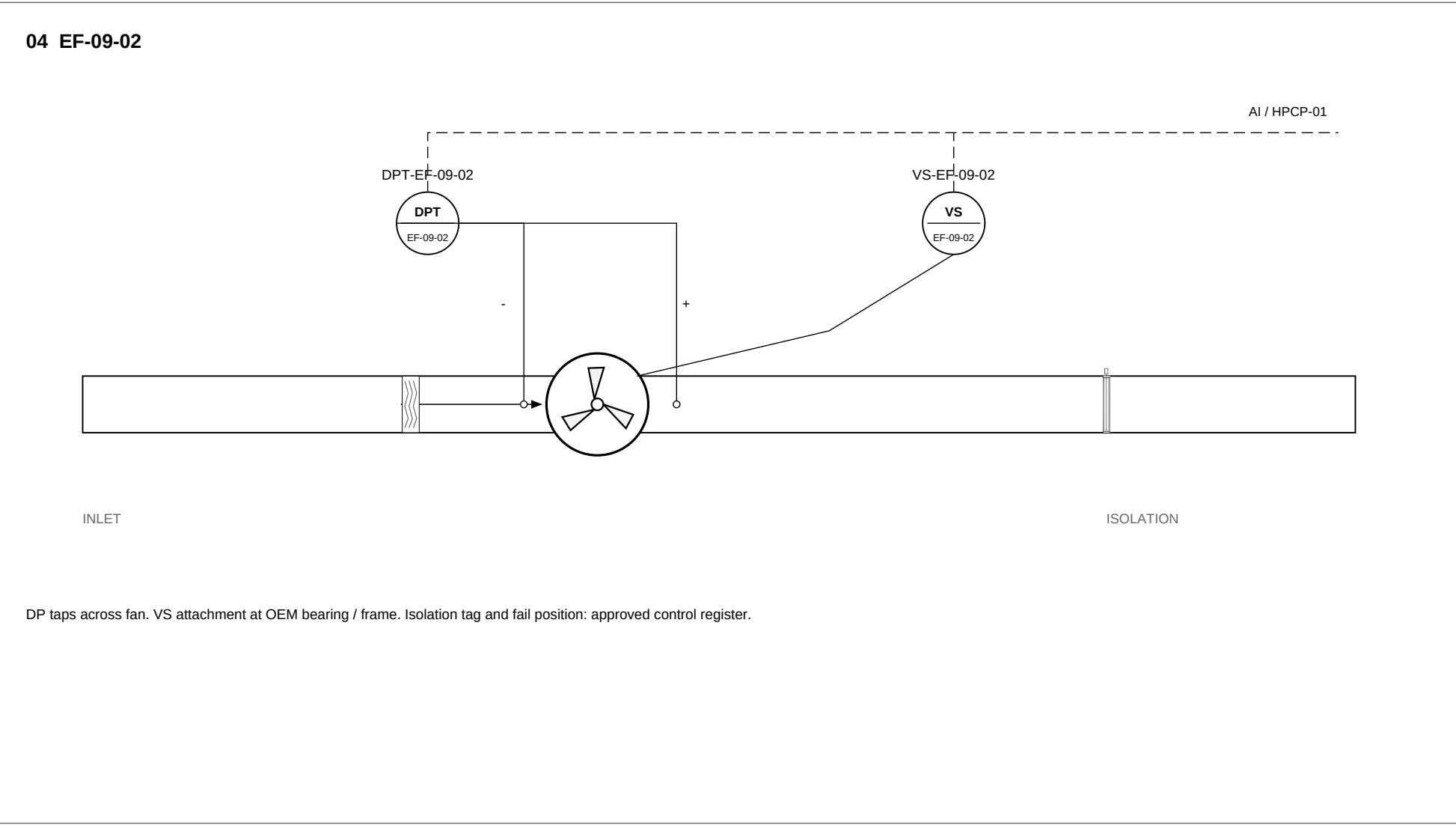
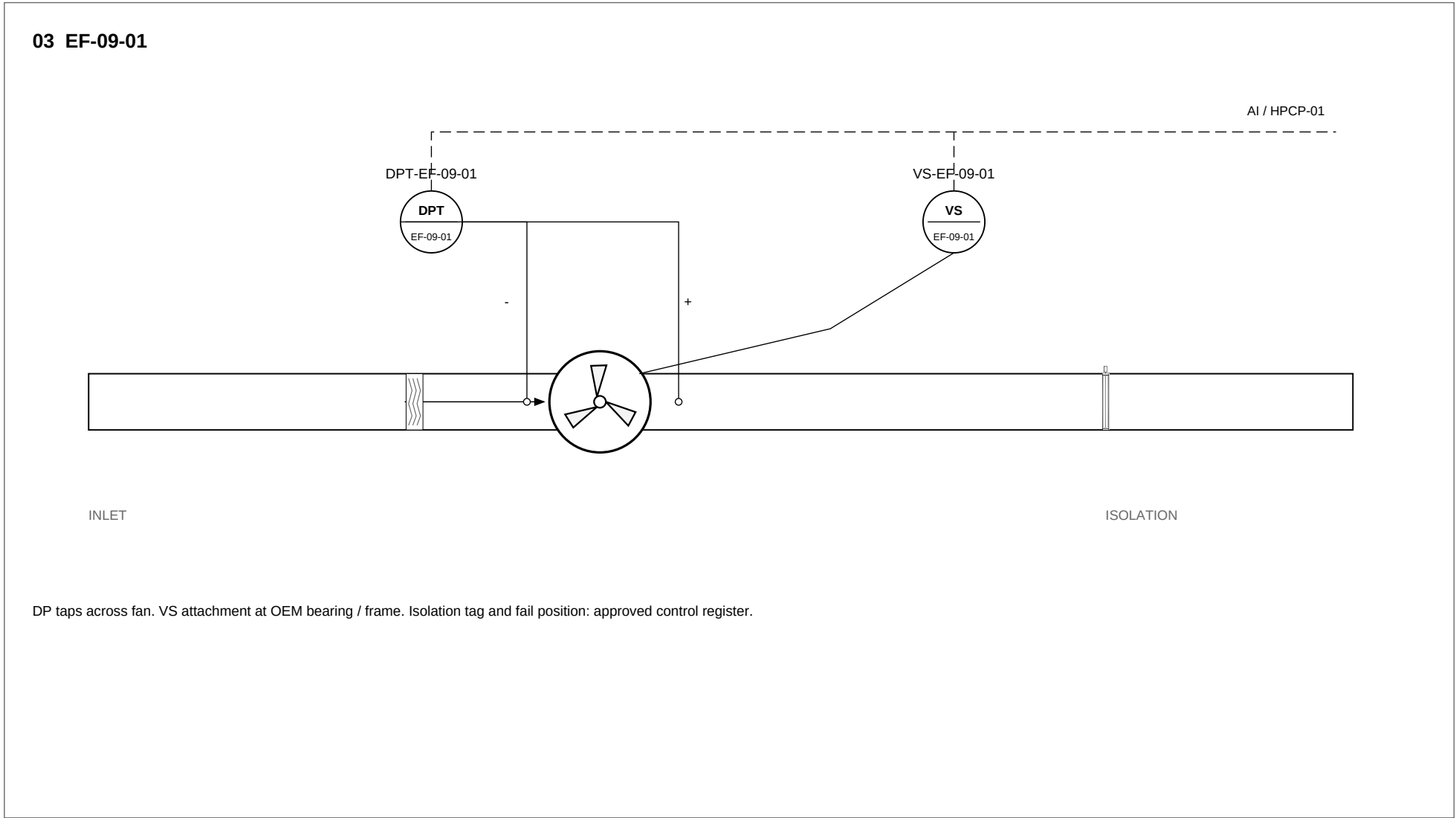
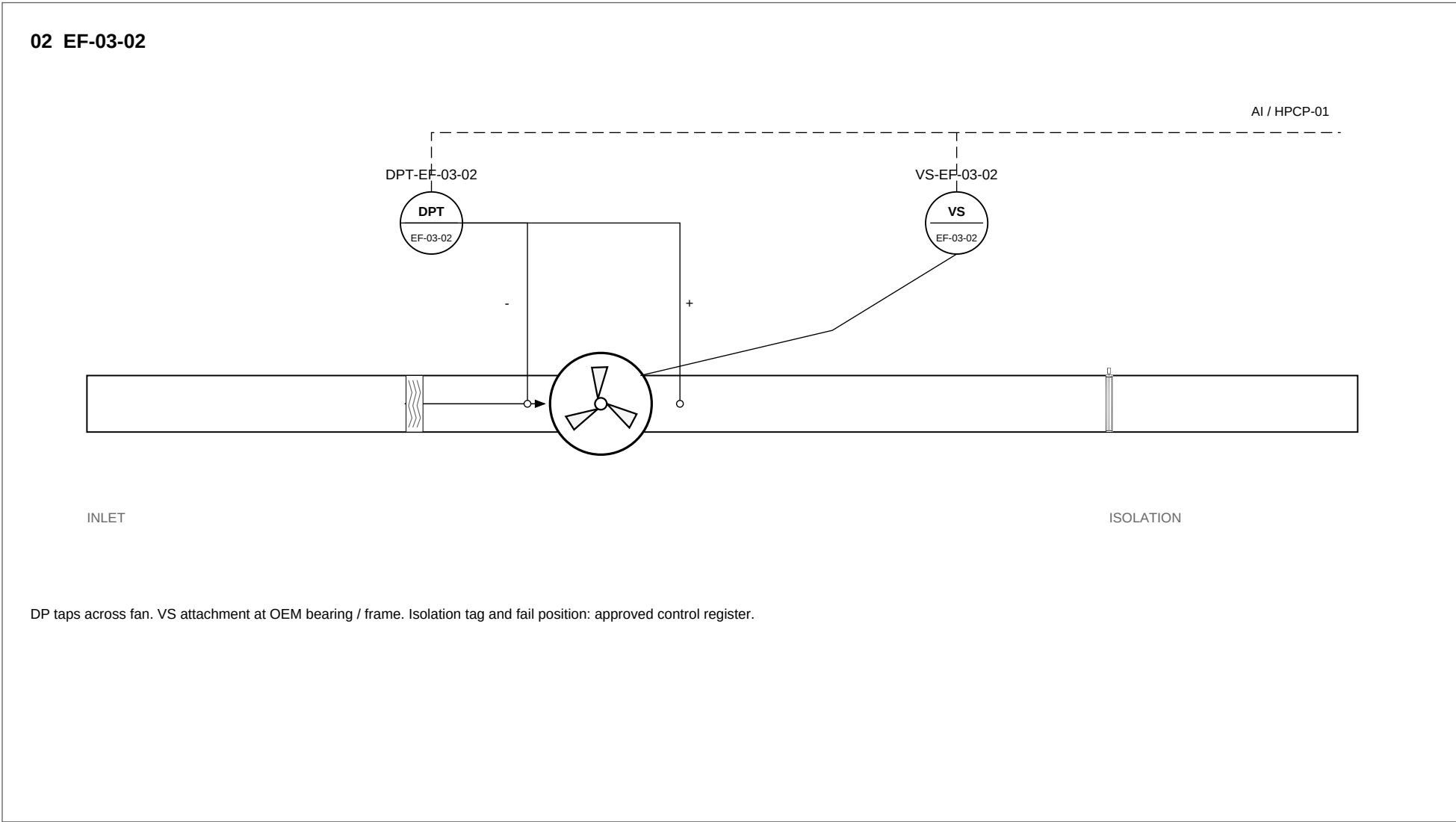
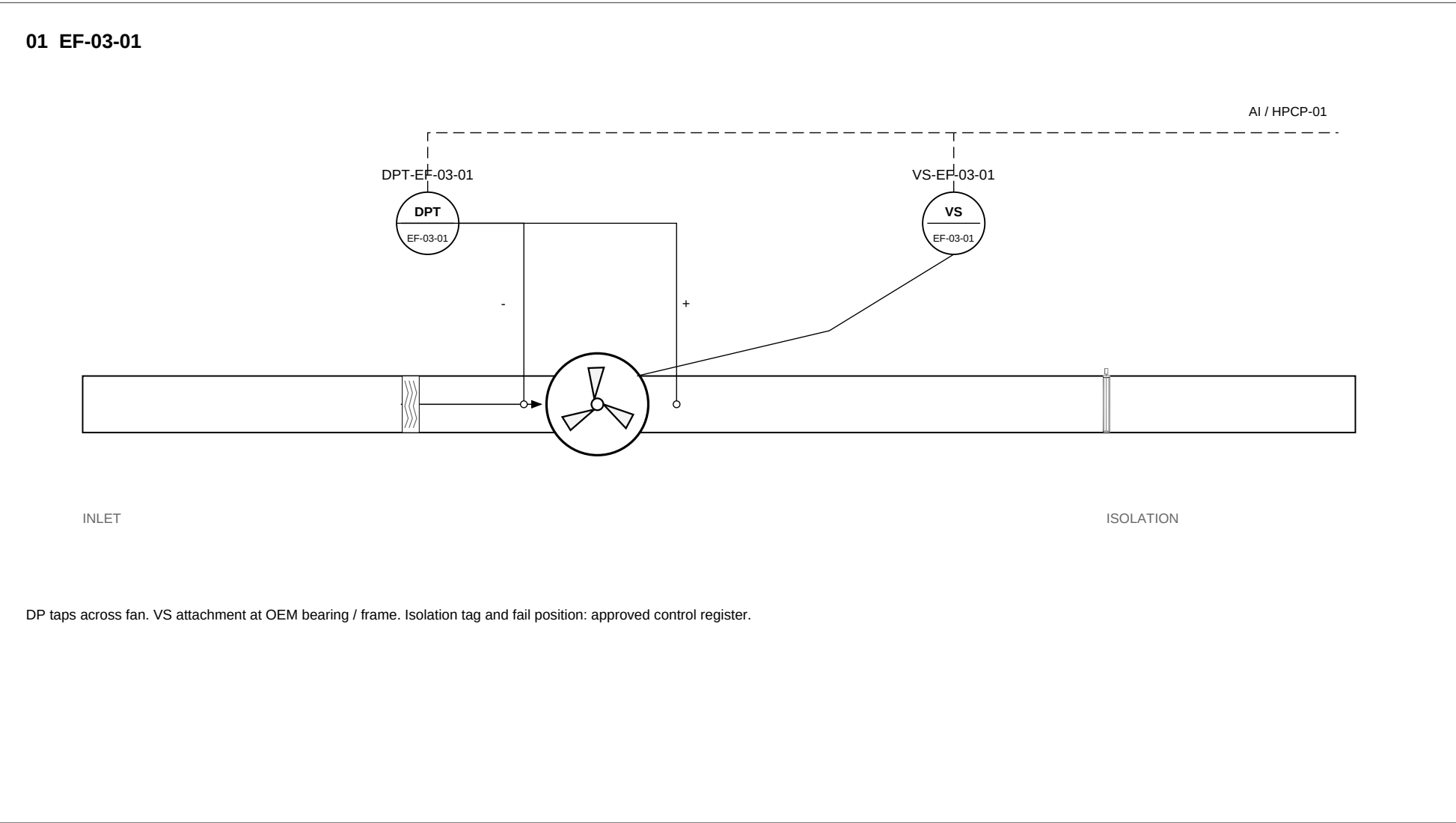
SCALE: NTS AT A1

REVISION

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NTS functional arrangements. Basement equipment positions are not shown on the Administration plan. CAG fan duty / quantity, tap locations, alarm limits and vibration mountings require OEM / I&C confirmation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

EXTRACT AND BASEMENT FAN INSTRUMENTATION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-013

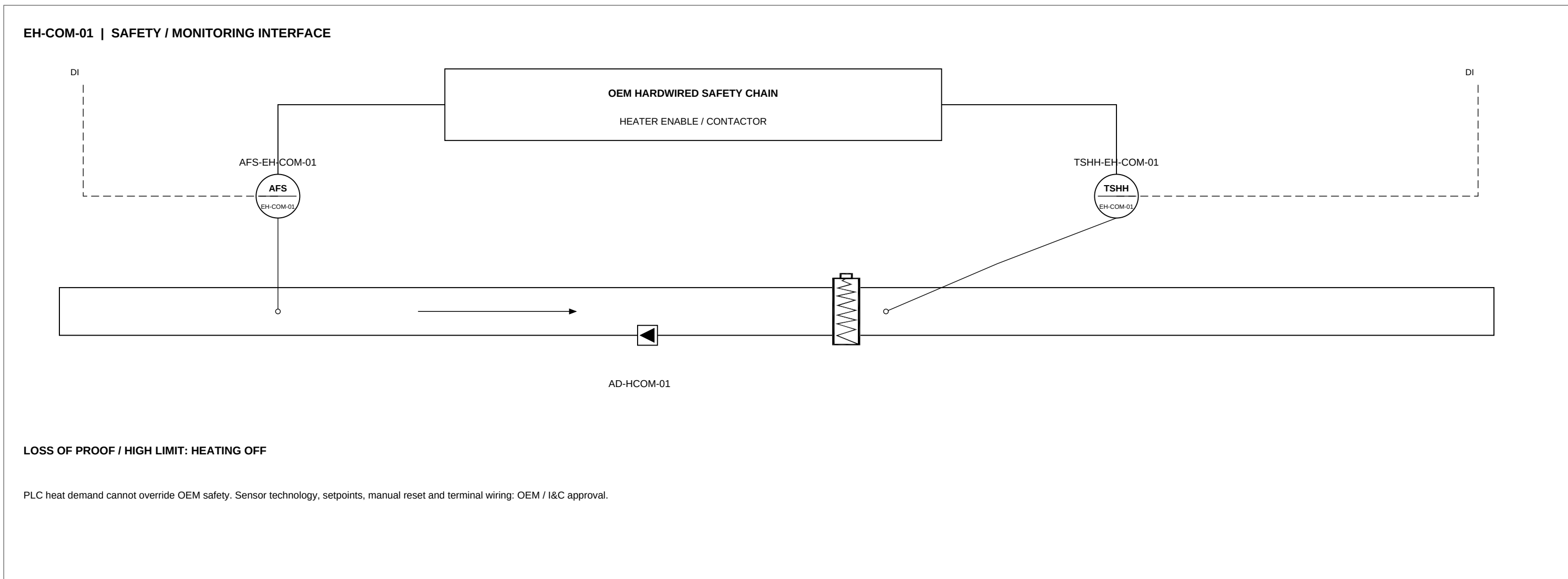
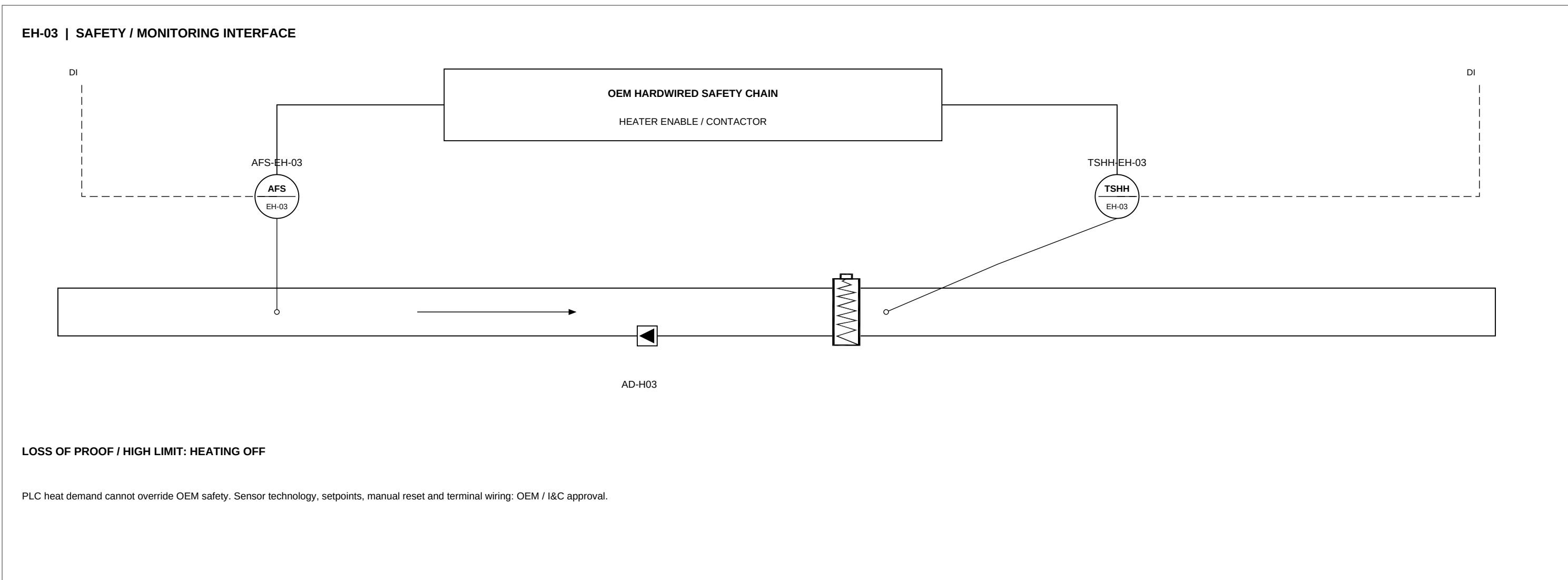
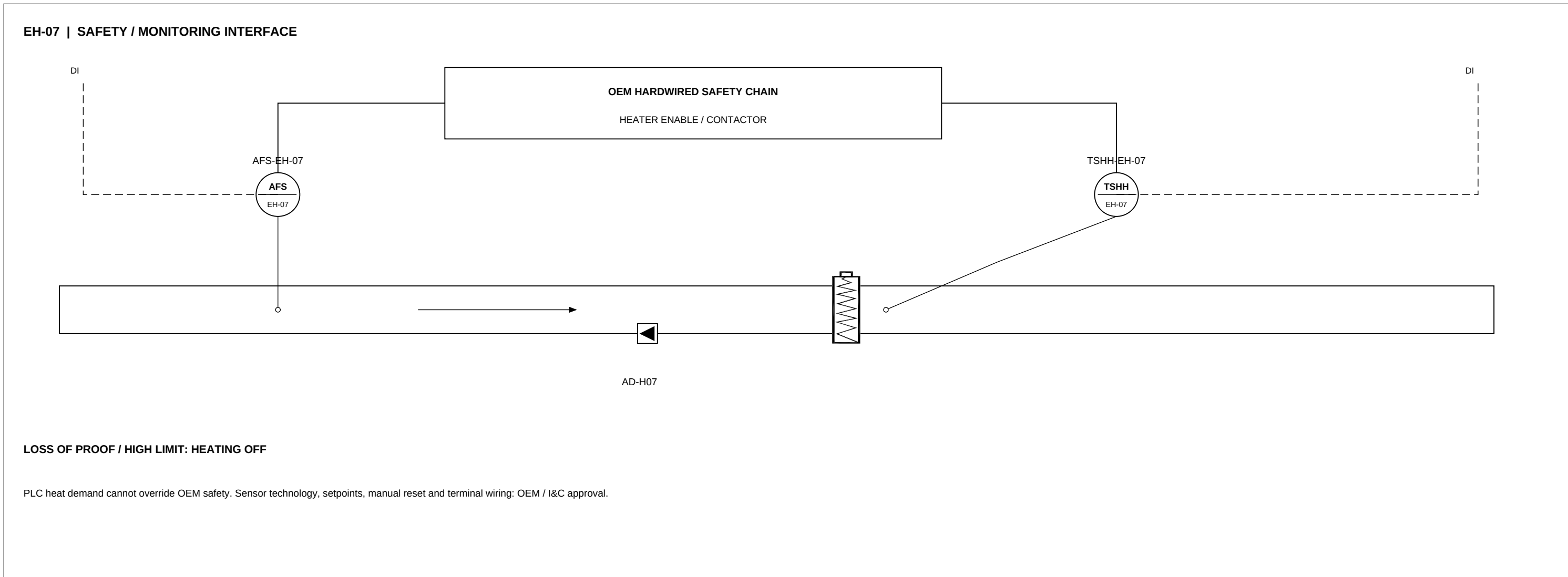
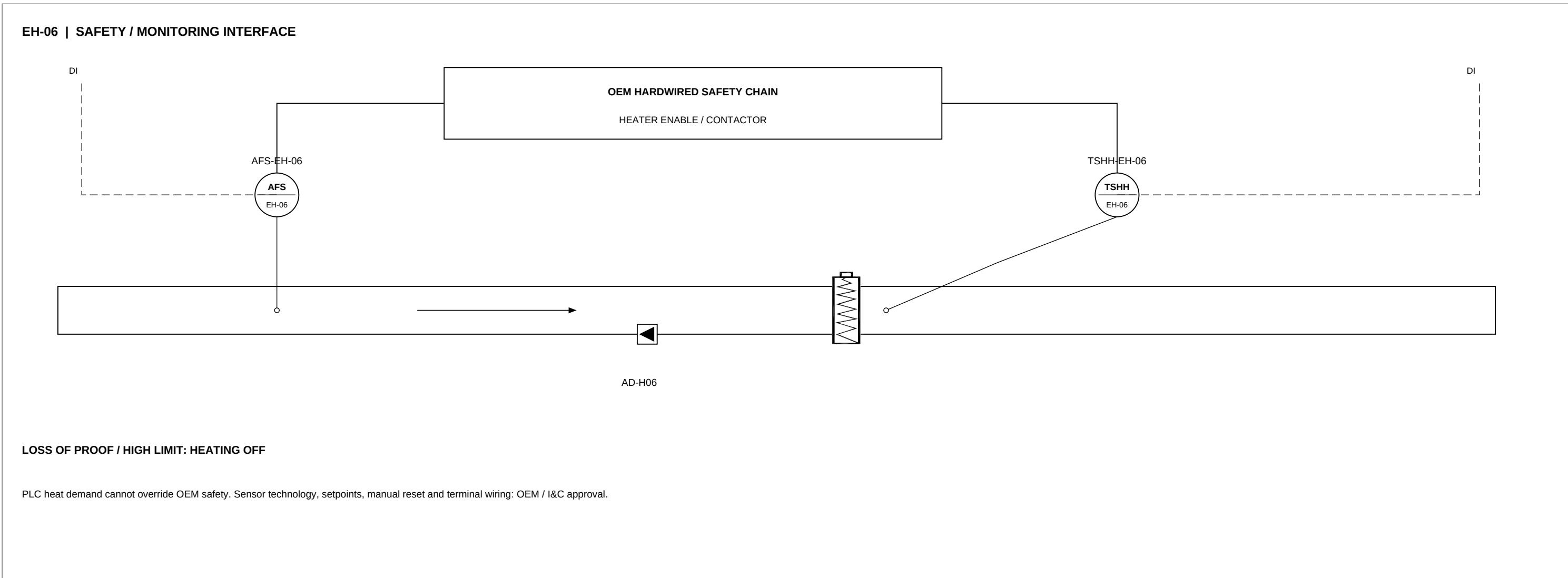
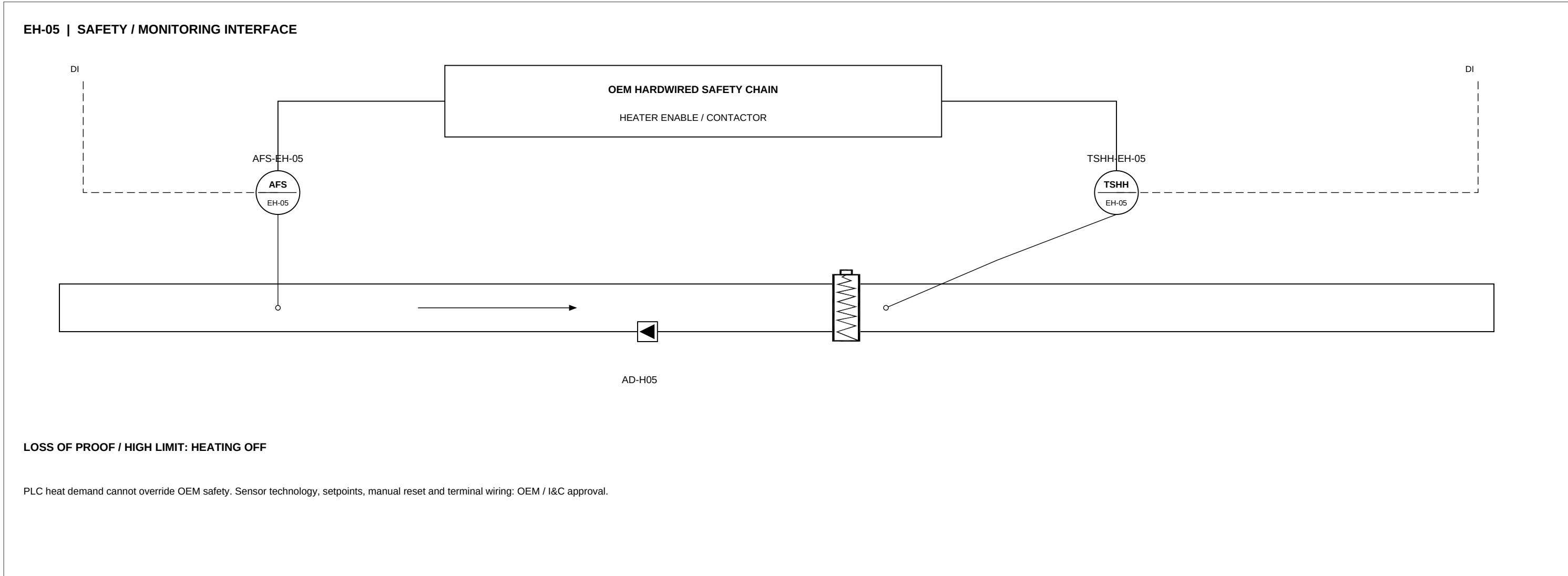
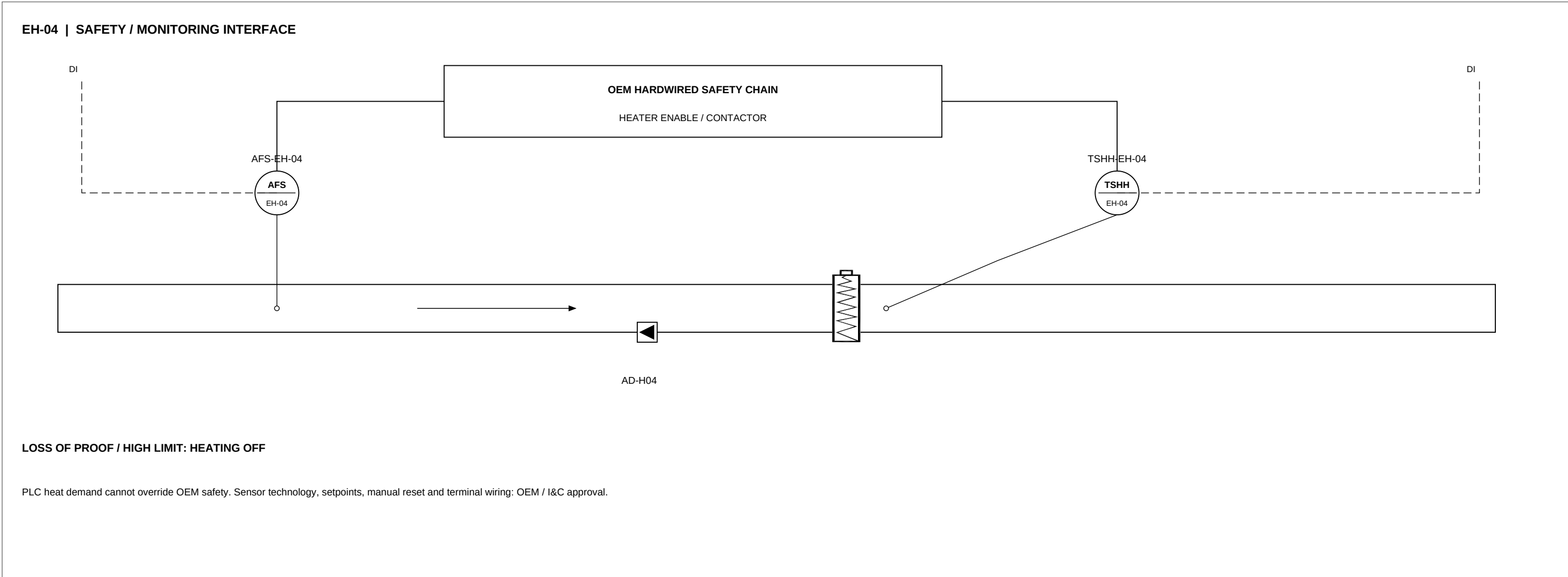
SCALE: NTS AT A1

REVISION

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SHEET 13 OF 18



Functional interfaces, not certified wiring diagrams. Six airflow-proof DI and six high-limit DI tags are retained. Protection circuits must be independently implemented and approved by the heater manufacturer

01 INSTRUMENT / SIGNAL CONVENTIONS

- TT

Temperature transmitter
- RHT

Humidity transmitter
- DPT

Differential pressure transmitter
- FIT

Airflow transmitter
- POT

Remote filter DP transmitter
- POI

Local differential pressure indication
- VS

Vibration sensing
- AFS

Airflow proving switch
- TSHH

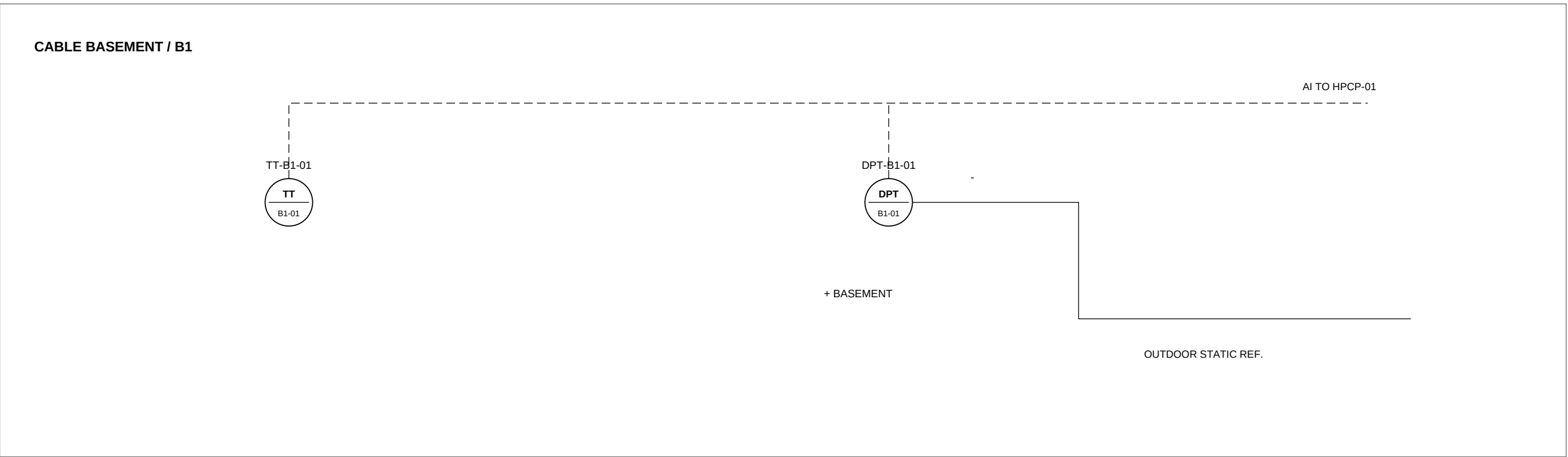
Heater high-temperature limit

———— Sensing / physical interface (route schematic)

----- Electrical / communications signal (schematic)

AI / AO / DI / DO are point classes, not selected signal ranges. 4-20 mA, contacts, network protocol, cable types and terminal numbers are not inferred.

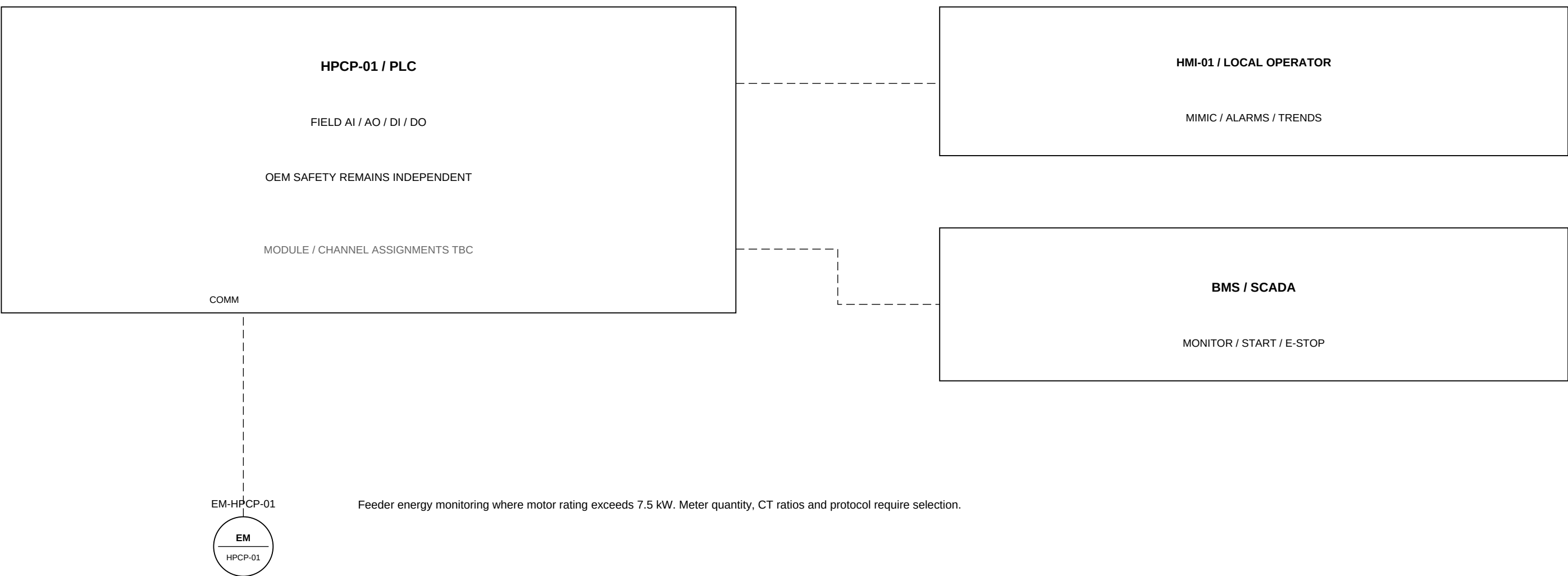
02 BASEMENT FIELD SENSING - PROPOSED



Basement temperature, volume, fan duties and actual pressure target remain open. Fan instrumentation is fully tagged on D-013; no basement architectural layout was supplied.

All controls shown are functional design intent. Approved OEM wiring and project I&C schedules govern implementation. D-016 provides an instrument-to-drawing cross-reference for every one of the 85 scheduled instruments.

03 HPCP / HMI / BMS INTERFACE



Feeder energy monitoring where motor rating exceeds 7.5 kW. Meter quantity, CT ratios and protocol require selection.

04 POINT CLASSES / MINIMUM SPARE

Class	Scheduled	20% spare	Min. channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM interfaces are separate from hardwired channels. Local PDI instruments have no PLC channel. Additional proposals are identified in the instrument index, D-016.

05 FIRE / STARTUP DESIGN INTENT

Confirmed fire: stop every HVAC fan and heater. Restart after fire ESD requires local manual reset. Damper fire positions are governed by the approved cause-and-effect, not this drawing.

Normal operation: two duty 50% PAUs with an available alternate. Prove selected air paths before starting fans. Duty rotation, changeover, timing and final permissives require approved OEM logic.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-015

SCALE: NTS AT A1

REVISION

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ADMINISTRATION BUILDING - COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

Instrument tag	I/O	Service / location	Status	D-sheet
TT-GF-02	AI	GF-02 Elec. panel	Required	007
RHT-GF-02	AI	GF-02 Elec. panel	Required	007
DPT-GF-02	AI	GF-02 Elec. panel	Proposal	007
TT-GF-04	AI	GF-04 Meeting	Required	007
RHT-GF-04	AI	GF-04 Meeting	Required	007
DPT-GF-04	AI	GF-04 Meeting	Proposal	007
TSHH-EH-04	DI	EH-04	OEM safety	014
TT-GF-08	AI	GF-08 Telecom	Required	007
RHT-GF-08	AI	GF-08 Telecom	Required	007
DPT-GF-08	AI	GF-08 Telecom	Proposal	007
TT-GF-05	AI	GF-05 Operator	Required	007
RHT-GF-05	AI	GF-05 Operator	Required	007
DPT-GF-05	AI	GF-05 Operator	Proposal	007
TSHH-EH-05	DI	EH-05	OEM safety	014
TT-GF-06	AI	GF-06 Corridor	Required	007
RHT-GF-06	AI	GF-06 Corridor	Required	007
DPT-GF-06	AI	GF-06 Corridor	Required	007
TSHH-EH-06	DI	EH-06	OEM safety	014
TT-GF-09	AI	GF-09 Toilets	Required	007
RHT-GF-09	AI	GF-09 Toilets	Required	007
DPT-GF-09	AI	GF-09 Toilets	Proposal	007
TT-GF-07	AI	GF-07 Rest room	Required	007
RHT-GF-07	AI	GF-07 Rest room	Required	007
DPT-GF-07	AI	GF-07 Rest room	Proposal	007
TSHH-EH-07	DI	EH-07	OEM safety	014
TT-GF-03	AI	GF-03 Kitchen	Required	007
RHT-GF-03	AI	GF-03 Kitchen	Required	007
DPT-GF-03	AI	GF-03 Kitchen	Proposal	007
TSHH-EH-03	DI	EH-03	OEM safety	014
TT-GF-01	AI	GF-01 Clean agent	Required	007
RHT-GF-01	AI	GF-01 Clean agent	Proposal	007
DPT-GF-01	AI	GF-01 Clean agent	Proposal	007
TT-SA-01	AI	Common supply	Required	007 / 008 / 012
TT-RA-01	AI	Common return	Required	007 / 012
RHT-RA-01	AI	Common return	Required	007 / 012
FIT-SA-01	AI	Common SA	Required	007 / 008 / 012
FIT-RA-01	AI	Common RA	Required	007 / 012
PDT-OA-01	AI	F-OA-01	Proposal	012
PDI-OA-01	LOCAL	F-OA-01	Proposal	012
TSHH-EH-COM-01	DI	EH-COM-01	OEM safety	014
PDT-F1-A	AI	PAU-A filter 1	Required	012
PDI-F1-A	LOCAL	PAU-A filter 1	Required	012
PDT-F2-A	AI	PAU-A filter 2	Required	012

Instrument tag	I/O	Service / location	Status	D-sheet
PDI-F2-A	LOCAL	PAU-A filter 2	Required	012
DPT-SF-A	AI	PAU-A fan	Required	012
VS-PAU-A	AI	PAU-A frame / fan bearing	Required	012
PDT-F1-B	AI	PAU-B filter 1	Required	012
PDI-F1-B	LOCAL	PAU-B filter 1	Required	012
PDT-F2-B	AI	PAU-B filter 2	Required	012
PDI-F2-B	LOCAL	PAU-B filter 2	Required	012
DPT-SF-B	AI	PAU-B fan	Required	012
VS-PAU-B	AI	PAU-B frame / fan bearing	Required	012
PDT-F1-C	AI	PAU-C filter 1	Required	012
PDI-F1-C	LOCAL	PAU-C filter 1	Required	012
PDT-F2-C	AI	PAU-C filter 2	Required	012
PDI-F2-C	LOCAL	PAU-C filter 2	Required	012
DPT-SF-C	AI	PAU-C fan	Required	012
VS-PAU-C	AI	PAU-C frame / fan bearing	Required	012
DPT-EF-03-01	AI	EF-03-01	Proposal	013
VS-EF-03-01	AI	EF-03-01	Required	013
DPT-EF-03-02	AI	EF-03-02	Proposal	013
VS-EF-03-02	AI	EF-03-02	Required	013
DPT-EF-09-01	AI	EF-09-01	Proposal	013
VS-EF-09-01	AI	EF-09-01	Required	013
DPT-EF-09-02	AI	EF-09-02	Proposal	013
VS-EF-09-02	AI	EF-09-02	Required	013
DPT-EF-01-01	AI	EF-01-01	Proposal	013
VS-EF-01-01	AI	EF-01-01	Required	013
DPT-SF-B1-01	AI	SF-B1-01	Proposal	013
VS-SF-B1-01	AI	SF-B1-01	Required	013
DPT-SF-B1-02	AI	SF-B1-02	Proposal	013
VS-SF-B1-02	AI	SF-B1-02	Required	013
DPT-EF-B1-01	AI	EF-B1-01	Proposal	013
VS-EF-B1-01	AI	EF-B1-01	Required	013
DPT-EF-B1-02	AI	EF-B1-02	Proposal	013
VS-EF-B1-02	AI	EF-B1-02	Required	013
TT-B1-01	AI	Cable basement	Required	015
DPT-B1-01	AI	Cable basement	Proposal	015
EM-HPCP-01	COMM	HPCP feeders >7.5 kW	Required*	015
AFS-EH-04	DI	Electric heater EH-04	Proposal	014
AFS-EH-05	DI	Electric heater EH-05	Proposal	014
AFS-EH-06	DI	Electric heater EH-06	Proposal	014
AFS-EH-07	DI	Electric heater EH-07	Proposal	014
AFS-EH-03	DI	Electric heater EH-03	Proposal	014
AFS-EH-COM-01	DI	Electric heater EH-COM-01	Proposal	014

85 unique instruments; each has a tagged graphic and drawing reference. Proposal / OEM safety / conditional metering statuses are retained. D-sheet references are ADM-HVAC-D-nnn; local PDI instruments are not wired AI points.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-016

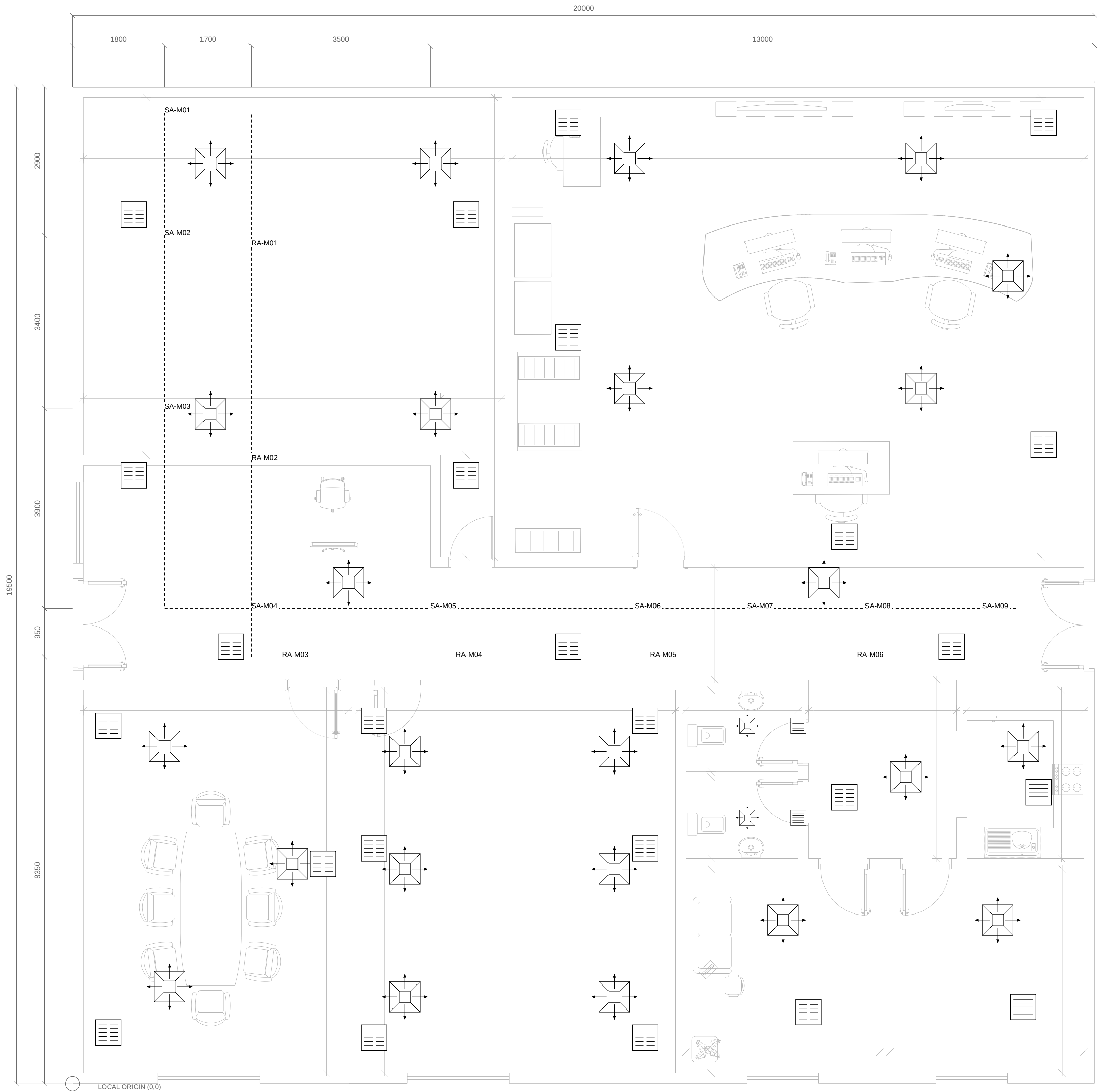
SCALE: NTS AT A1

REVISION

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Dimension chains are plan centerline offsets and architectural extents only. They are not approved fabrication, support or vertical-clearance dimensions. Terminal faces retain scheduled dimensions, final catalogue free areas and performance remain pending.

SOURCE PLAN COORDINATES / DATUM

Dimensions and coordinates are derived from the supplied architectural plan. Local (0,0) is a drawing origin, not a surveyed site benchmark. Confirm all setting-out against the coordinated ceiling and architecture.

GF	Terminal X range mm	Terminal Y range mm
02	1200 - 7700	11900 - 18000
05	9700 - 19000	10700 - 18800
04	700 - 4900	1000 - 7000
08	5900 - 11200	900 - 7100
06	3100 - 17200	5600 - 9800
07	13900 - 14400	1400 - 3200
09	13200 - 14200	5200 - 7000
03	18600 - 18900	5700 - 6600
01	18100 - 18600	1500 - 3200

Individual terminal coordinates: Terminals register. Main centerlines: Duct_sections register. Native CAD dimensions are provided for the known overall extents and main offsets.

Duct BOD, ceiling datum, riser/drop levels, insulation thickness and crossing clearances remain TBC. The plan does not establish a safe vertical separation or a developed fabrication length.

INSPECTION / COORDINATION NOTES

1. Verify dimensions on site; do not scale NTS details.
2. Match device tags to registers before procurement.
3. Reserve access to dampers, heaters and filters.
4. Obtain selected-fitting losses and OEM duties.
5. Close pressure / outdoor-air design holds.

ADMINISTRATION BUILDING - DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES

DETAILED DRAWING INDEX

Drawing number	Drawing title	Scale
ADM-HVAC-D-001	COORDINATED DUCT PLAN	1:50
ADM-HVAC-D-002	SUPPLY AIR DUCT PLAN	1:50
ADM-HVAC-D-003	RETURN AIR AND EXTRACT DUCT PLAN	1:50
ADM-HVAC-D-004	NORTH ROOMS - DUCT ENLARGEMENT	1:35
ADM-HVAC-D-005	SOUTH-WEST ROOMS - DUCT ENLARGEMENT	1:25
ADM-HVAC-D-006	SOUTH-EAST ROOMS - DUCT ENLARGEMENT	1:25
ADM-HVAC-D-007	ROOM INSTRUMENTATION AND CONTROL LOCATIONS	1:50
ADM-HVAC-D-008	PAU BANK AND INDOOR RISER SECTION	NTS
ADM-HVAC-D-009	DUCT SYMBOLS AND INSTALLATION DETAILS	NTS
ADM-HVAC-D-010	DUCT AND TERMINAL SCHEDULES	NTS
ADM-HVAC-D-011	TAPERED FITTINGS AND MAINTENANCE ACCESS	NTS
ADM-HVAC-D-012	PLANT INSTRUMENTATION AND HEATER PROTECTION	NTS
ADM-HVAC-D-013	EXTRACT AND BASEMENT FAN INSTRUMENTATION	NTS
ADM-HVAC-D-014	ELECTRIC HEATER PROTECTION AND CONTROL	NTS
ADM-HVAC-D-015	FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES	NTS
ADM-HVAC-D-016	COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES	NTS
ADM-HVAC-D-017	PLAN DIMENSIONS AND TERMINAL SETTING-OUT	1:50
ADM-HVAC-D-018	DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES	NTS

DESIGN BASIS / OPEN ENGINEERING ITEMS

Supply 3697.80 L/s; assumed return 3006.68 L/s. Pressure targets +50 Pa normal rooms, +25 Pa kitchen / toilets / clean-agent; actual balance unverified.

OA reference 420.70 L/s is below 691.12 L/s implied makeup. Toilet net SA-EA is -18.69 L/s; kitchen net is zero. Neither proves the +25 Pa target. Resolve transfer air / extract balance and outdoor-air compliance.

Fan ESP / curves, fitting K, vertical lengths, levels, support and insulation, diffuser throw / noise, access envelopes and I&C settings remain pending.

ISSUE STATUS: FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION. Drawn / checked / approved identities and authority are pending. Reference styling does not establish design compliance or professional sign-off.

REVISION 09 - SCOPE OF ISSUE

Drafting revision: layer / lineweight hierarchy, readable tag conventions, native plan dimensions and instrument reference register.

Complete tagged fan / heater protection and basement / energy interfaces are added. Existing 85 instrument requirements, 114 duct sizes and scheduled room airflows are retained.

Reference D-113391-ADDC-AES-ME-AC-001_REV.1 is a drafting benchmark; its building duties and instrument requirements are not substituted for this Administration design.

DRAWING / COORDINATION CONVENTIONS

All duct sizes: clear internal width x height in mm; D denotes round diameter. Flows: L/s; velocities: m/s. Add approved insulation outside clear sizes.

Supply / return crossings without a registered junction are independent paths. Plan white masks provide graphic separation only; elevations are not certified.

MFD* graphic corresponds to conditional FSD-prefixed register tags. Approval of rated boundaries governs quantity, rating and fail positions.

TT / RHT / DPT room locations are proposed. Pressure references are outdoors. Dashed I&C lines are functional signals, not installed cable routes.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-018

SCALE: NTS AT A1

REVISION

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